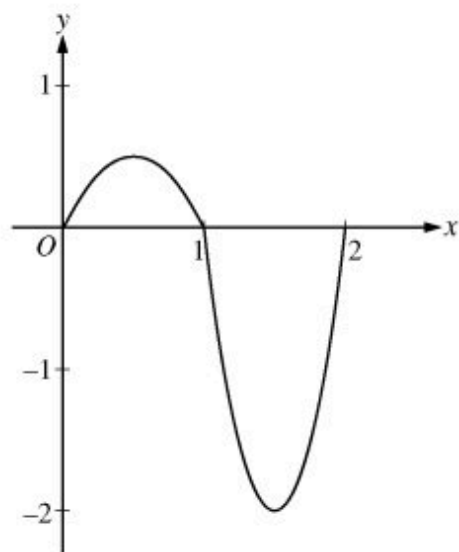


5-5

1.

Graph of f'

The figure above shows the graph of f' , the derivative of a function f , for $0 \leq x \leq 2$. What is the value of x at which the absolute minimum of f occurs?

- (A) 0
(B) $\frac{1}{2}$
(C) 1
(D) $\frac{3}{2}$
(E) 2



2. Let f be the function given by $f(x) = x^3 - 6x^2 - 15x$. What is the maximum value of f on the interval $[0, 6]$?

- (A) 0
(B) 5
(C) 6
(D) 8



5-5

3.  The number of people who have entered a museum on a certain day is modeled by a function $f(t)$, where t is measured in hours since the museum opened that day. The number of people who have left the museum since it opened that same day is modeled by a function $g(t)$. If $f'(t) = 380(1.02^t)$ and $g'(t) = 240 + 240 \sin\left(\frac{\pi(t-4)}{12}\right)$, at what time t , for $1 \leq t \leq 11$, is the number of people in the museum at a maximum?

(A) 1

(B) 7.888

(C) 9.446

(D) 10.974

(E) 11

4. The function g is given by $g(x) = 4x^3 + 3x^2 - 6x + 1$. What is the absolute minimum value of g on the closed interval $[-2, 1]$?

(A) -7

(B) $-\frac{3}{4}$

(C) 0

(D) 2

(E) 6

5-5

5. Let $g(x) = xe^{-x} + be^{-x}$, where b is a positive constant.

For what positive value of b does g have an absolute maximum at $x = \frac{2}{3}$? Justify your answer.

Part B

2 points are earned for $g'(x)$

1 point is earned for $g'(\frac{2}{3}) = 0$ for b

1 point is earned for justification

$$g'(x) = e^{-x} - xe^{-x} - be^{-x} = (1 - x - b)e^{-x}$$

$$g'(\frac{2}{3}) = (\frac{1}{3} - b)e^{2/3} = 0 \Rightarrow b = \frac{1}{3}$$

$$\text{When } b = \frac{1}{3}, g'(x) = (\frac{2}{3} - x)e^{-x}.$$

For $x < \frac{2}{3}$, $g'(x) > 0$ and for $x > \frac{2}{3}$, $g'(x) < 0$.

Therefore, when $b = \frac{1}{3}$, g has an absolute maximum

at $x = \frac{2}{3}$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for $g'(\frac{2}{3}) = 0$ for b

1 point is earned for justification

$$g'(x) = e^{-x} - xe^{-x} - be^{-x} = (1 - x - b)e^{-x}$$

$$g'(\frac{2}{3}) = (\frac{1}{3} - b)e^{2/3} = 0 \Rightarrow b = \frac{1}{3}$$

$$\text{When } b = \frac{1}{3}, g'(x) = (\frac{2}{3} - x)e^{-x}.$$

5-5

For $x < \frac{2}{3}$, $g'(x) > 0$ and for $x > \frac{2}{3}$, $g'(x) < 0$.

Therefore, when $b = \frac{1}{3}$, g has an absolute maximum

at $x = \frac{2}{3}$.

6.  Let f be the function with first derivative defined by $f'(x) = \sin(x^3)$ for $0 \leq x \leq 2$. At what value of x does f attain its maximum value on the closed interval $0 \leq x \leq 2$?

(A) 0

(B) 1.162

(C) 1.465

(D) 1.845

(E) 2



5-5

7. The following are related to this scenario:

A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \leq t \leq 12$. At time $t=0$, when the sale begins, there are no shoppers in the store.

 Using the models, $S(t)$ and $L(t) = -80 + \frac{4400}{t^2 - 14t + 55}$, find the time t , $0 \leq t \leq 12$, at which the number of shoppers in the store is the greatest. Justify your answer.

Part D

The response can earn up to 3 points:

1 point: For consideration $S'(t) - L(t) = 0$

1 point: For the correct answer

1 point: For justification of the answer.

$$N(t) = S(t) - \int_0^t L(x) dx$$

$$N'(t) = S'(t) - L(t) = 0 \text{ when } t = 5.45066$$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.45$ hours.



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: For consideration $S'(t) - L(t) = 0$

1 point: For the correct answer

1 point: For justification of the answer

5-5

$$N(t) = S(t) - \int_0^t L(x)dx$$

$$N'(t) = S'(t) - L(t) = 0 \text{ when } t = 5.45066$$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.45$ hours.

5-5

8. The penguin population on an island is modeled by a differentiable function P of time t , where $P(t)$ is the number of penguins and t is measured in years for $0 \leq t \leq 40$. There are 100,000 penguins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$$B(t) = 1000e^{0.06t} \text{ penguins per year}$$

and the death rate for the penguins on the island is modeled by

$$D(t) = 250e^{0.1t} \text{ penguins per year.}$$



To the nearest whole number, find the absolute minimum penguin population and the absolute maximum penguin population on the island for $0 \leq t \leq 40$. Show the analysis that leads to your answer.

Part D

The response can earn up to 4 points:

1 point: For the expression $B(t) - D(t) = 0$

$$B(t) - D(t) = 0$$

1 point: For the solution for t

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point: For the minimum value

1 point: For the maximum value

$$P(0) = 100000$$

$$P(A) = 100000 + \int_0^A (B(t) - D(t)) dt = 139166.667$$

$$P(40) = 133058$$

The minimum population is 100,000 and the maximum population is 139,167 penguins.



0	1	2	3	4
---	---	---	---	---

The response earns all four of the following points:

1 point: For the expression $B(t) - D(t) = 0$

$$B(t) - D(t) = 0$$

1 point: For the solution for t

5-5

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point: For the minimum value

1 point: For the maximum value

$$P(0) = 100000$$

$$P(A) = 100000 + \int_0^A (B(t) - D(t))dt = 139166.667$$

$$P(40) = 133058$$

The minimum population is 100,000 and the maximum population is 139,167 penguins.

5-5

9. Particle X moves along the positive x -axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$.
At what time $t \geq 0$ is particle X farthest to the left? Justify your answer.

Part B

The response can earn up to 3 points:

1 point: Consideration of $x'(t) = 0$

1 point: Identification of $t = \frac{6}{5}$

1 point: Correct answer with reason

$$x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0, t = \frac{6}{5}$$

Since $x'(t) < 0$, for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$, the particle is farthest to the left at time $t = \frac{6}{5}$



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: Consideration of $x'(t) = 0$

1 point: Identification of $t = \frac{6}{5}$

1 point: Correct answer with reason

$$x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0, t = \frac{6}{5}$$

Since $x'(t) < 0$, for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$, the particle is farthest to the left at time $t = \frac{6}{5}$

5-5

5-5

10. The rate at which raw sewage enters a treatment tank is given by $E(t) = 850 + 715 \cos\left(\frac{\pi t^2}{9}\right)$ gallons per hour for $0 \leq t \leq 4$ hours. Treated sewage is removed from the tank at the constant rate of 645 gallons per hour. The treatment tank is empty at time $t = 0$.



For $0 \leq t \leq 4$ at what time t is the amount of sewage in the treatment tank greatest? To the nearest gallon, what is the maximum amount of sewage in the tank? Justify your answers.

Part B

One point is earned for sets $E(t) = 645$

One point is earned for identifies $t = 2.309$ as a candidate

One point is earned for the amount of sewage at $t = 2.309$

One point is earned for the conclusion

Let $S(t)$ be the amount of sewage in the treatment tank at time t . Then $S'(t) = E(t) - 645$ and $S'(t) = 0$ when

$E(t) = 645$. On the interval $0 \leq t \leq 4$, $E(t) = 645$ when $t = 2.309$ and $t = 3.559$

The amount of sewage in the treatment tank is greatest at $t = 2.309$ hours. At that time, the amount of sewage in the tank, rounded to the nearest gallon, is 1637 gallons.

t (hours)	amount of sewage in treatment tank
0	0
2.309	$\int_0^{2.309} E(t) dt - 645(2.309) = 1637.178$
3.559	$\int_0^{3.559} E(t) dt - 645(3.559) = 1228.520$
4	$3981.022 - 645(4) = 1401.022$



0	1	2	3	4
---	---	---	---	---

The student earns all of the following points:

5-5

One point is earned for sets $E(t) = 645$

One point is earned for identifies $t = 2.309$ as a candidate

One point is earned for the amount of sewage at $t = 2.309$

One point is earned for the conclusion

Let $S(t)$ be the amount of sewage in the treatment tank at time t . Then $S'(t) = E(t) - 645$ and $S'(t) = 0$ when

$E(t) = 645$. On the interval $0 \leq t \leq 4$, $E(t) = 645$ when $t = 2.309$ and $t = 3.559$

The amount of sewage in the treatment tank is greatest at $t = 2.309$ hours. At that time, the amount of sewage in the tank, rounded to the nearest gallon, is 1637 gallons.

t (hours)	amount of sewage in treatment tank
0	0
2.309	$\int_0^{2.309} E(t) dt - 645(2.309) = 1637.178$
3.559	$\int_0^{3.559} E(t) dt - 645(3.559) = 1228.520$
4	$3981.022 - 645(4) = 1401.022$

11.  The derivative of the function f is given by $f'(x) = e^{-x} \cos(x^2)$, for all real numbers x . What is the minimum value of $f(x)$ for $-1 \leq x \leq 1$?

(A) $f(-1)$

(B) $f(-0.762)$

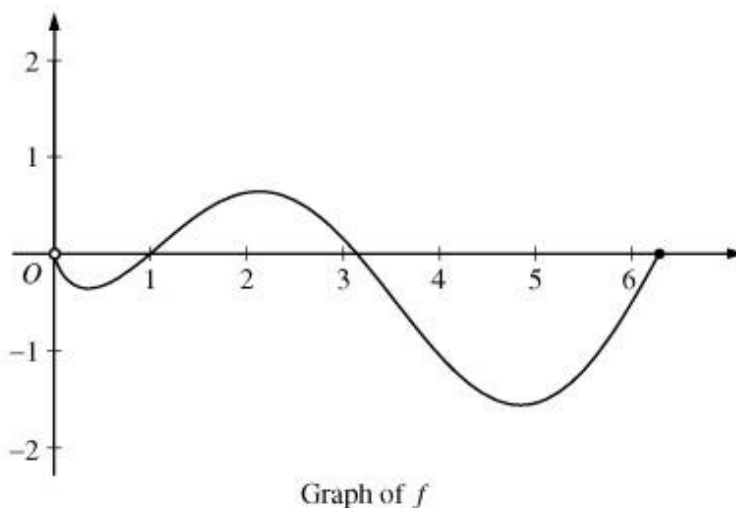
(C) $f(1)$

(D) There is no minimum value of $f(x)$ for $-1 \leq x \leq 1$.



5-5

12.



Let f be the function given by $f(x) = (\ln x)(\sin x)$. The figure above shows the graph of f for $0 < x \leq 2\pi$.

The function g is defined by $g(x) = \int_1^x f(t) dt$ for $0 < x \leq 2\pi$.

For $0 < x \leq 2\pi$ find the value of x at which g has an absolute minimum. Justify your answer.

Part C

1 point is earned for identifies 1 and 2π as candidates

or indicates that the graph of g decreases, increases, then decreases

1 point is earned for justifies $g(2\pi) < g(1)$

1 point is earned for the answer

For $0 < x < 2\pi$, $g'(x) = f(x) = 0$ when $x = 1, \pi$. $g' = f$ changes from negative to positive only at $x = 1$. The absolute minimum must occur at $x = 1$ or at the right endpoint. Since $g(1) = 0$ and

$$g(2\pi) = \int_1^{2\pi} f(t) dt = \int_1^{\pi} f(t) dt + \int_{\pi}^{2\pi} f(t) dt < 0$$

by comparison of the two areas, the absolute minimum occurs at $x = 2\pi$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifies 1 and 2π as candidates

or indicates that the graph of g decreases, increases, then decreases

5-5

1 point is earned for justifies $g(2\pi) < g(1)$

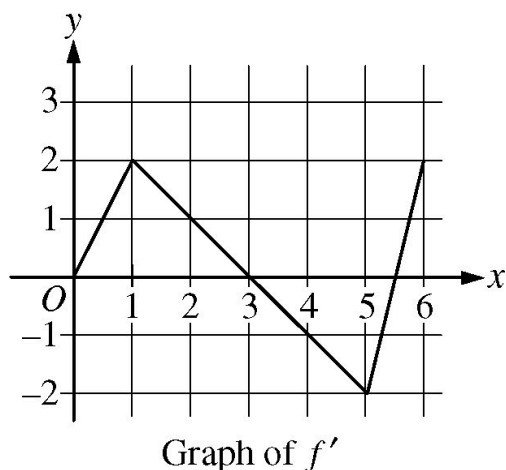
1 point is earned for the answer

For $0 < x < 2\pi$, $g'(x) = f(x) = 0$ when $x = 1, \pi$. $g' = f$ changes from negative to positive only at $x = 1$. The absolute minimum must occur at $x = 1$ or at the right endpoint. Since $g(1) = 0$ and

$$g(2\pi) = \int_1^{2\pi} f(t) dt = \int_1^{\pi} f(t) + \int_{\pi}^{2\pi} f(t) dt < 0$$

by comparison of the two areas, the absolute minimum occurs at $x = 2\pi$.

13.



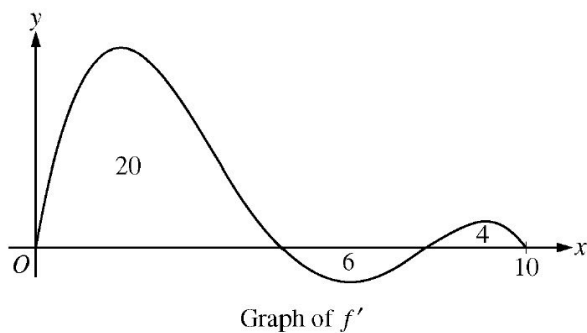
For $0 \leq x \leq 6$, the graph of f' , the derivative of f , is piecewise linear as shown above. If $f(0) = 1$, what is the maximum value of f on the interval?

- (A) 1
- (B) 1.5
- (C) 2
- (D) 4
- (E) 6



5-5

14.



- (A) 16
- (B) 20
- (C) 22
- (D) 30
- (E) 32



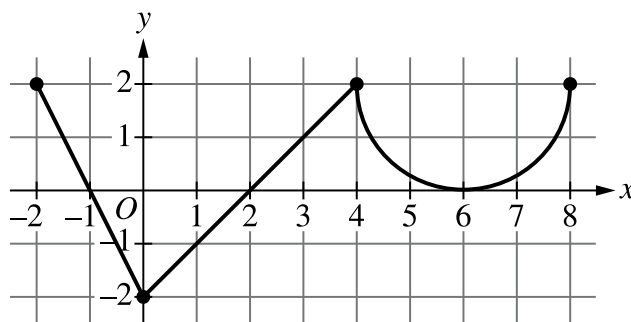
5-5

15. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The function f is defined on the closed interval $[-2, 8]$ and satisfies $f(2) = 1$. The graph of f' , the derivative of f , consists of two line segments and a semicircle, as shown in the figure.

- Does f have a relative minimum, a relative maximum, or neither at $x = 6$? Give a reason for your answer.
- On what open intervals, if any, is the graph of f concave down? Give a reason for your answer.
- Find the value of $\lim_{x \rightarrow 2} \frac{6f(x) - 3x}{x^2 - 5x + 6}$, or show that it does not exist. Justify your answer.
- Find the absolute minimum value of f on the closed interval $[-2, 8]$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that declares $f'(x)$ does not change sign at $x = 6$, so neither, is sufficient to earn the point.

A response does not have to present intervals on which $f'(x)$ is positive or negative, but if any are given, they must be correct. Any presented intervals may include none, one, or both endpoints.

A response that declares $f'(x) > 0$ before and after $x = 6$ does not earn the point.



0

1

5-5

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$f'(x) > 0$ on $(2, 6)$ and $f'(x) > 0$ on $(6, 8)$.

$f'(x)$ does not change sign at $x = 6$, so there is neither a relative maximum nor a relative minimum at this location.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only by an answer of $(-2, 0)$ and $(4, 6)$, or these intervals including one or both endpoints.

A response must earn the first point to be eligible for second point.

To earn the second point a response must correctly discuss the behavior of f' or the slopes of f' .

Special case: A response that presents exactly one of the two correct intervals with a correct reason earns 1 out of 2 points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Intervals

☐ Reason

Solution:

The graph of f is concave down on $(-2, 0)$ and $(4, 6)$ because f' is decreasing on these intervals.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the presentation of two separate limits for the numerator and denominator.

A response that presents a limit explicitly equal to $\frac{0}{0}$ does not earn the first point.

The second point is earned by applying L'Hospital's Rule, that is, by presenting at least one correct derivative in the limit of a ratio of derivatives.

5-5

The third point is earned for the correct answer with supporting work.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator
- ☐ Uses L'Hospital's Rule
- ☐ Answer

Solution:

Because f is differentiable at $x = 2$, f is continuous at $x = 2$, so $\lim_{x \rightarrow 2} f(x) = f(2) = 1$.

$$\lim_{x \rightarrow 2} (6f(x) - 3x) = 6 \cdot 1 - 3 \cdot 2 = 0$$

$$\lim_{x \rightarrow 2} (x^2 - 5x + 6) = 0$$

Because $\lim_{x \rightarrow 2} \frac{6f(x) - 3x}{x^2 - 5x + 6}$ is of indeterminate form $\frac{0}{0}$, L'Hospital's Rule can be applied.

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 2} \frac{6f(x) - 3x}{x^2 - 5x + 6} = \lim_{x \rightarrow 2} \frac{6f'(x) - 3}{2x - 5} = \frac{6 \cdot 0 - 3}{2 \cdot 2 - 5} = 3.$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must state $f' = 0$ or equivalent. Listing the zeros of f' is not sufficient.

A response that presents any error in evaluating f at any critical point or endpoint will not earn the justification point.

A response need not present the value of $f(-1)$ provided $x = -1$ is eliminated because it is the location of a local maximum. A response need not present the value of $f(6)$ provided $x = 6$ is eliminated by reference to part (a) or eliminated through analysis.

A response need not present the value of $f(8)$ provided there is a presentation that argues $f'(x) \geq 0$ for $x > 2$ and, therefore, $f(8) > f(2)$.

A response that does not consider both endpoints does not earn the justification point.

5-5

The answer point is earned only for indicating that the minimum value is 1. It is not earned for noting that the minimum occurs at $x = 2$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $f'(x) = 0$
- ☐ Justification
- ☐ Answer

Solution:

$$f'(x) = 0 \Rightarrow x = -1, x = 2, x = 6$$

The function f is continuous on $[-2, 8]$, so the candidates for the location of an absolute minimum for f are $x = -2$, $x = -1$, $x = 2$, $x = 6$, and $x = 8$.

x	$f(x)$
-2	3
-1	4
2	1
6	$7 - \pi$
8	$11 - 2\pi$

The absolute minimum value of f is $f(2) = 1$.

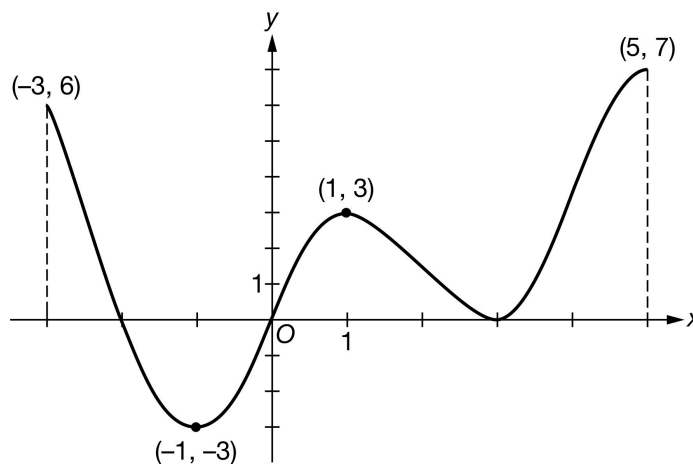
5-5

16. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The figure above shows the graph of g' , the derivative of a twice-differentiable function g , on the closed interval $-3 \leq x \leq 5$. The graph of g' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of g' on the intervals $[-3, -2]$, $[-2, 0]$, $[0, 3]$, and $[3, 5]$ are 3, 4, 5, and 7, respectively. The function g is defined for all real numbers and satisfies $g(1) = 3$ and $g(5) = 13$.

- Find the x -coordinate of each critical point of g on the open interval $-3 < x < 5$.
- Identify the critical point from part (a) which corresponds to a relative maximum of g . Give a reason for your answer.
- Find the value of g at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute maximum value of g on $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals is g both decreasing and concave up? Give a reason for your answer.
- Explain why there must be a value of a , for $1 < a < 5$, such that $g(a) = 10$.
- Find $\lim_{x \rightarrow 5^-} \frac{g(x) - 13}{3x - 15}$. Give a reason for your answer.
- The function r is defined by $r(x) = (g(x))^3$. Find $r'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-5

To earn this point, the response must present all three critical values: $x = -2$, $x = 0$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -2$, $x = 0$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-2, 0)$, $(0, 0)$, $(3, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = 0$ at $x = -2$, $x = 0$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local maximum as -2 and give the correct reason of “ g' changes from positive to negative” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ g switches from increasing to decreasing at $x = -2$ ”
- Saying that “ $g'(-2)$ changes signs from positive to negative”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of g' , the one given in the stem of the problem.

A response indicating that “ $g'(x)$ is positive for $x < -2$ and negative for $x > -2$ ” is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.

5-5



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative maximum value of g occurs where $g'(x)$ changes from positive to negative values.

$x = -2$ is the only location on $-3 < x < 5$ for which this is true, so $x = -2$ is the only location of a relative maximum value of g on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = 0$, or $x = 3$ is selected as the location of the local maximum in part (b). Note that $g(-3) = 2$, $g(0) = 1$, and $g(3) = 6$. No other x -coordinates will be read for consistency.

The response $g(-2) = g(5) - (-4 + 5 + 7) = 5$ earns both points.

The response of both $13 - 8 = 5$ and $g(-2) = 5$ does not earn the first point, but earns the second point, provided no other work is presented.

The string of equalities $g(-2) = g(5) - \int_{-2}^5 g'(x) dx = 13 - 8 = 5$ earns both points.

Special Case: If in part (b), a response picks $x = 1$ or $x = 5$ as the location of the maximum, then there is no work to do in part (c). This response will earn zero points.

5-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $g(-2) = g(5) + \int_5^{-2} g'(x) dx$

☐ Answer

Solution:

$$\begin{aligned}
 g(-2) &= g(5) + \int_5^{-2} g'(x) dx \\
 &= g(5) - \int_{-2}^5 g'(x) dx = 13 - (7 + 5 - 4) = 5
 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = -2$, the response must either state its y -value or reference it as a maximum.

For the endpoints, the y -values must be presented but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 1$), and thus the third point cannot be earned.

It is possible to not refer to $g(-3) = 2$ by indicating that g' is positive on the interval $(-3, -2)$, thus eliminating 2 as a possible absolute maximum.

A complete list of given or computable values for $g(x)$ is given in the table below.

5-5

x	$g(x)$
-3	2
-2	5
0	1
1	3
3	6
5	13



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = -2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$$g(-3) = g(5) + \int_5^{-3} g'(x) dx$$

$$= g(5) - \int_{-3}^5 g'(x) dx = 13 - (7 + 5 - 4 + 3) = 2$$

$$g(-2) = 5$$

$$g(5) = 13$$

So the absolute maximum value of g on $-3 \leq x \leq 5$ is $g(5) = 13$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-1, 0]$, $[-1, 0)$, $(-1, 0]$, $(-1, 0)$ all earn the first point.

The reason “ g' is increasing” can be expressed by either “ g'' is positive” or “the slope of g' is positive.”

We will overlook the false implication that $g' < 0$ on $[-1, 0]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[-1, 0]$.

5-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where g' is negative.

g is concave up where g' is increasing.

g is both decreasing and concave up on the open interval $(-1, 0)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that g is continuous because g is differentiable or g is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[1, 5]$, or the point is not earned.

The y -values of g need to be mentioned by either referring to the numerical values of 3 and 13, or mentioning $g(1)$ and $g(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since g is differentiable it’s continuous and 10 is between 3 and 13 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

5-5

g is differentiable. $\Rightarrow g$ is continuous.

$$g(1) = 3 < 10 < 13 = g(5)$$

By the Intermediate Value Theorem, there must be a value of a , for $1 < a < 5$, such that $g(a) = 10$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of g' or g .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $g(5) - 13 = 0$ and $3 \cdot 5 - 15 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $g(5) = 13$ and $3 \cdot 5 = 15$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g(5)-13}{3 \cdot 5-15} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g'(x)}{3} = \frac{7}{3}$, will earn the second point, but not the third point; note that even just one of two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{0}{0}$ and

$\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g'(x)}{3}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{7}{3}$, or its numeric equivalent.

5-5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ both g' and g are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} g(x) = g(5)$ and $\lim_{x \rightarrow 5^-} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5^-} (g(x) - 13) = g(5) - 13 = 13 - 13 = 0$$

$$\lim_{x \rightarrow 5^-} (3x - 15) = 3 \cdot 5 - 15 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{g(x) - 13}{3x - 15} &= \lim_{x \rightarrow 5^-} \frac{g'(x)}{3} \\ &= \frac{g'(5)}{3} = \frac{7}{3} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $3 \cdot (g(x))^2$ or $3 \cdot (g(1))^2$.

The chain rule point can be earned by seeing an expression involving g but not g' times $g'(1)$.

All three points will be awarded for seeing only $3 \cdot 3^2 \cdot 3$, but $3 \cdot 9 \cdot 3$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

5-5

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$r'(x) = 3(g(x))^2 \cdot g'(x)$$

$$r'(1) = 3(g(1))^2 \cdot g'(1) = 3 \cdot 3^2 \cdot 3 = 81$$

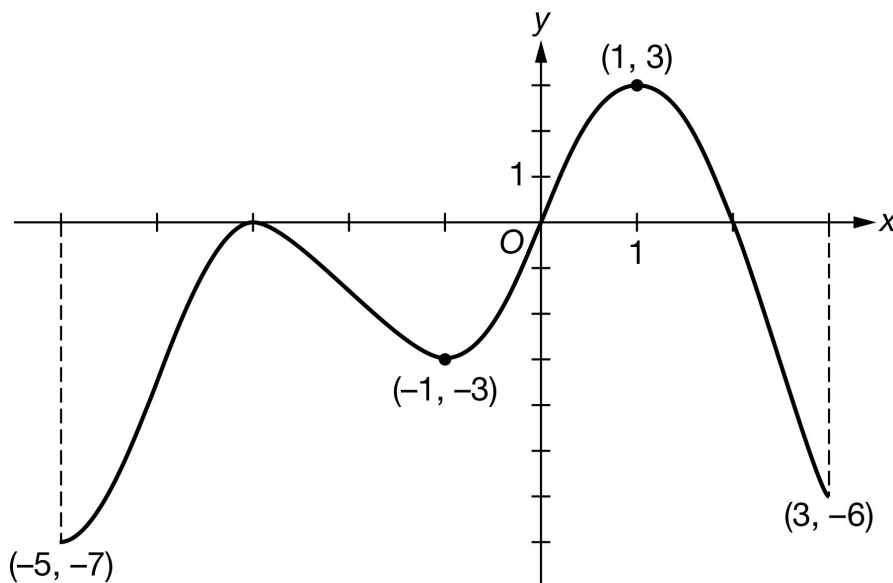
5-5

17. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The figure above shows the graph of h' , the derivative of a twice-differentiable function h , on the closed interval $-5 \leq x \leq 3$. The graph of h' has horizontal tangent lines at $x = -3$, $x = -1$, and $x = 1$. The areas of the regions bounded by the x -axis and the graph of h' on the intervals $[-5, -3]$, $[-3, 0]$, $[0, 2]$, and $[2, 3]$ are 7, 5, 4, and 3, respectively. The function h is defined for all real numbers and satisfies $h(1) = 12$ and $h(3) = 11$.

- Find the x -coordinate of each critical point of h on the open interval $-5 < x < 3$.
- Identify the critical point from part (a) which corresponds to a relative maximum of h . Give a reason for your answer.
- Find the value of h at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute maximum value of h on $-5 \leq x \leq 3$. Justify your answer.
- On what open intervals is h both increasing and concave down? Give a reason for your answer.
- Explain why there must be a value of b , for $1 < b < 3$, such that $h(b) = 11.5$.
- Find $\lim_{x \rightarrow 3^-} \frac{h(x) - 11}{x^2 - 9}$. Give a reason for your answer.
- The function p is defined by $p(x) = (h(x))^2$. Find $p'(1)$. Show the computations that lead to your answer.

5-5

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = 0$, and $x = 2$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5, 3)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5, 3)$. The point is still earned provided the three critical points, $x = -3$, $x = 0$, and $x = 2$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(0, 0)$, $(2, 0)$ we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = 0$ at $x = -3$, $x = 0$, and $x = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local maximum as 2 and give the correct reason of “ h' changes from positive to negative” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ h switches from increasing to decreasing at $x = 2$ ”
- Saying that “ $h'(2)$ changes signs from positive to negative”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of h' , the one given in the stem of the problem.

A response indicating that “ $h'(x)$ is positive for $x < 2$ and negative for $x > 2$ ” is allowed to give this statement as a reason to earn the point.

5-5

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative maximum value of h occurs where $h'(x)$ changes from positive to negative values.

$x = 2$ is the only location on $-3 < x < 5$ for which this is true, so $x = 2$ is the only location of a relative maximum value of h on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -5$, $x = -3$, or $x = 0$ is selected as the location of the local maximum in part (b). Note that $h(-5) = 22$, $h(-3) = 15$, and $h(0) = 10$. No other x -coordinates will be read for consistency.

The response $h(2) = h(3) - (-3) = 14$ earns both points.

The response of both $11 - (-3) = 14$ and $h(2) = 14$ does not earn the first point, but earns the second point, provided no other work is presented.

Special Case: If a response uses $\int_1^2 h'(x) dx = 2$ then

5-5

$h(2) = h(1) + \int_1^2 h'(x) dx = 12 + 2 = 14$ earns both points, because it's possible to find $\int_1^2 h'(x) dx = 2$ using the areas and initial conditions given.

any response using perceived symmetry about $x = 1$ can earn at most the first point, for example

$$h(2) = h(1) + \int_1^2 h'(x) dx = 12 + \frac{1}{2} \cdot 4 = 14 \text{ earns just the first point.}$$

Special Case: If in part (b), a response picks $x = 1$ or $x = 3$ as the location of the maximum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $h(2) = h(3) + \int_3^2 h'(x) dx$

☐ Answer

Solution:

$$\begin{aligned} h(2) &= h(3) + \int_3^2 h'(x) dx \\ &= h(3) - \int_2^3 h'(x) dx = 11 - (-3) = 14 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x - value.

For $x = 2$, the response must either state its y -value or reference it as a maximum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 1$), and thus the third point cannot be earned.

5-5

It is possible to not refer to $h(3) = 11$ by indicating that h' is negative on the interval $(2, 3)$, thus eliminating 11 as a possible absolute maximum.

A complete list of given or computable values for $h(x)$ is given in the table below

x	$h(x)$
-5	22
-3	15
0	10
1	12
2	14
3	11



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$$h(-5) = h(3) + \int_3^{-5} h'(x) dx$$

$$= h(3) - \int_{-5}^3 h'(x) dx = 11 - (-7 - 5 + 4 - 3) = 22$$

$$h(2) = 14$$

$$h(3) = 11$$

So the absolute maximum value of h on $-5 \leq x \leq 3$ is $h(-5) = 22$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[1, 2]$, $[1, 2)$, $(1, 2]$, and $(1, 2)$ all earn the first point.

The reason “ h' is decreasing” can be expressed by either “ h'' is negative” or “the slope of h' is negative.”

5-5

We will overlook the false implication that $h' > 0$ on $[1, 2]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[1, 2]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where h' is positive.

h is concave down where h' is decreasing.

h is both increasing and concave down on the open interval $(1, 2)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that h is continuous because h is differentiable or h is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[1, 3]$, or the point is not earned.

The y -values of h need to be mentioned by either referring to the numerical values of 11 and 12, or mentioning $h(1)$ and $h(3)$

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since h is differentiable it’s continuous and 11.5 is between 11 and 12 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation using Intermediate Value Theorem

5-5

Solution:

h is differentiable. $\Rightarrow h$ is continuous.

$$h(3) = 11 < 11.5 < 12 = h(1)$$

By the Intermediate Value Theorem, there must be a value of b , for $1 < b < 3$, such that $h(b) = 11.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of h' or h .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $h(3) - 11 = 0$ and $3^2 - 9 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $h(3) = 11$ and $3^2 = 9$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h(3)-11}{3^2-9} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{0}{0}$ earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h'(x)}{2x} = \frac{-6}{6}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{0}{0}$ and $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

5-5

The third point is earned only for the answer of $-\frac{6}{6}$ or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h is twice differentiable. $\Rightarrow$ both h' and h are continuous. $\Rightarrow \lim_{x \rightarrow 3^-} h(x) = h(3)$ and $\lim_{x \rightarrow 3^-} h'(x) = h'(3)$.

$$\lim_{x \rightarrow 3^-} (h(x) - 11) = h(3) - 11 = 11 - 11 = 0$$

$$\lim_{x \rightarrow 3^-} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3^-} \frac{h(x) - 11}{x^2 - 9} &= \lim_{x \rightarrow 3^-} \frac{h'(x)}{2x} \\ &= \frac{h'(3)}{2 \cdot 3} = \frac{-6}{6} = -1 \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $2 \cdot h(x)$ or $2 \cdot h(1)$.

The chain rule point can be earned by seeing an expression involving h but not h' times $h'(1)$.

All three points will be awarded for seeing only $2 \cdot 12 \cdot 3$, but $24 \cdot 3$, $2 \cdot 36$, or $6 \cdot 12$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

5-5

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$p'(x) = 2h(x) \cdot h'(x)$$

$$p'(1) = 2 \cdot h(1) \cdot h'(1) = 2 \cdot 12 \cdot 3 = 72$$

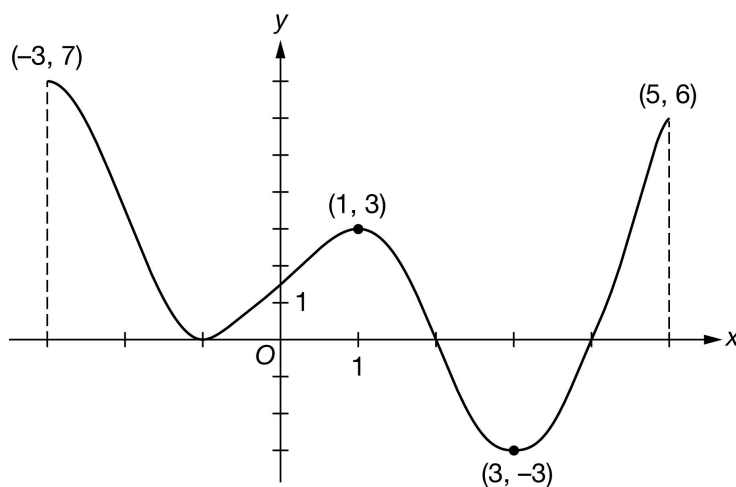
5-5

18. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The figure above shows the graph of g' , the derivative of a twice-differentiable function g , on the closed interval $-3 \leq x \leq 5$. The graph of g' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of g' on the intervals $[-3, -1]$, $[-1, 2]$, $[2, 4]$, and $[4, 5]$ are 7, 5, 4, and 3, respectively. The function g is defined for all real numbers and satisfies $g(3) = 8$ and $g(5) = 9$.

- Find the x -coordinate of each critical point of g on the open interval $-3 < x < 5$.
- Identify the critical point from part (a) which corresponds to a relative minimum of g . Give a reason for your answer.
- Find the value of g at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of g on $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals is g both decreasing and concave down? Give a reason for your answer.
- Explain why there must be a value of r , for $3 < r < 5$, such that $g(r) = 8.5$.
- Find $\lim_{x \rightarrow 5^-} \frac{g(x) - 9}{x^2 - 25}$. Give a reason for your answer.
- The function f is defined by $f(x) = (g(x))^3$. Find $f'(3)$. Show the computations that lead to your answer.

Part A

5-5

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -1$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(2, 0)$, $(4, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = 0$ at $x = -1$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 4 and give the correct reason of “ g' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ g switches from decreasing to increasing at $x = 4$ ”
- Saying that “ $g'(4)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of g' , the one given in the stem of the problem.

A response indicating that “ $g'(x)$ is negative for $x < 4$ and positive for $x > 4$ ” is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.

5-5



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 4$ is the only location on $-3 < x < 5$ for which this is true, so $x = 4$ is the only location of a relative minimum value of g on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = -1$, or $x = 2$ is selected as the location of the local minimum in part (b). Note that $g(-3) = -2$, $g(-1) = 5$, and $g(2) = 10$. No other x -coordinates will be read for consistency.

The response $g(4) = g(5) - 3 = 6$ earns both points.

The responses of both $9 - 3 = 6$ and $g(4) = 6$ do not earn the first point, but earns the second point, provided no other work is presented.

Special Case: If a response uses $\int_4^3 g'(x) dx = 2$ then

$g(4) = g(3) - \int_4^3 g'(x) dx = 8 - 2 = 6$ earns both points, because it's possible to find $\int_4^3 g'(x) dx = 2$ using the areas and initial conditions given.

5-5

any response using perceived symmetry about $x = 3$ can earn at most the first point, for example

$$g(4) = g(3) - \int_4^3 g'(x) dx = 8 - \frac{1}{2} \cdot 4 = 6 \text{ earns just the first point.}$$

Special Case: If in part (b), a response picks $x = 3$ or $x = 5$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $g(4) = g(5) + \int_5^4 g'(x) dx$

☐ Answer

Solution:

$$\begin{aligned} g(4) &= g(5) + \int_5^4 g'(x) dx \\ &= g(5) - \int_4^5 g'(x) dx = 9 - 3 = 6 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 4$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 3$), and thus the third point cannot be earned.

It is possible to not refer to $g(5) = 9$ by indicating that g' is positive on the interval $(4, 5)$, thus eliminating 9 as a possible absolute minimum.

A complete list of given or computable values for $g(x)$ is given in the table below.

5-5

x	$g(x)$
-3	-2
1	5
2	10
3	8
4	6
5	9



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 4$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$g(-3) = g(5) + \int_5^{-3} g'(x) dx$$

$$= g(5) - \int_{-3}^5 g'(x) dx = 9 - (7 + 5 - 4 + 3) = -2$$

$$g(4) = 6$$

$$g(5) = 9$$

So the absolute minimum value of g on $-3 \leq x \leq 5$ is $g(-3) = -2$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[2, 3]$, $[2, 3)$, $(2, 3]$, and $(2, 3)$ all earn the first point.

The reason “ g' is decreasing” can be expressed by either “ g'' is negative” or “the slope of g' is negative.”

We will overlook the false implication that $g' < 0$ on $[2, 3]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[2, 3]$.

5-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where g' is negative.

g is concave down where g' is decreasing.

g is both decreasing and concave down on the open interval $(2, 3)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that g is continuous because g is differentiable or g is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[3, 5]$, or the point is not earned.

The y -values of g need to be mentioned by either referring to the numerical values of 8 and 9, or mentioning $g(3)$ and $g(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since g is differentiable it’s continuous and 8.5 is between 8 and 9 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

5-5

g is differentiable. $\Rightarrow g$ is continuous.

$$g(3) = 8 < 8.5 < 9 = g(5)$$

By the Intermediate Value Theorem, there must be a value of r , for $3 < r < 5$, such that $g(r) = 8.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of g' or g .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $g(5) - 9 = 0$ and $5^2 - 25 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $g(5) = 9$ and $5^2 = 25$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g(5)-9}{5^2-25} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g'(x)}{2x} = \frac{3}{5}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{0}{0}$ and $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{3}{5}$, or its numeric equivalent.

5-5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ that both g' and g are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} g(x) = g(5)$ and $\lim_{x \rightarrow 5^-} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5^-} (g(x) - 9) = g(5) - 9 = 9 - 9 = 0$$

$$\lim_{x \rightarrow 5^-} (x^2 - 25) = 5^2 - 25 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{g(x) - 9}{x^2 - 25} &= \lim_{x \rightarrow 5^-} \frac{g'(x)}{2x} \\ &= \frac{g'(5)}{2 \cdot 5} = \frac{6}{10} = \frac{3}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $3 \cdot (g(x))^2$ or $3 \cdot (g(3))^2$.

The chain rule point can be earned by seeing an expression involving g but not g' times $g'(3)$.

All three points will be awarded for seeing only $3 \cdot 8^2 \cdot (-3)$, but $3 \cdot 64 \cdot (-3)$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

5-5

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$f'(x) = 3(g(x))^2 \cdot g'(x)$$

$$f'(3) = 3(g(3))^2 \cdot g'(3) = 3 \cdot 8^2 \cdot (-3) = -576$$

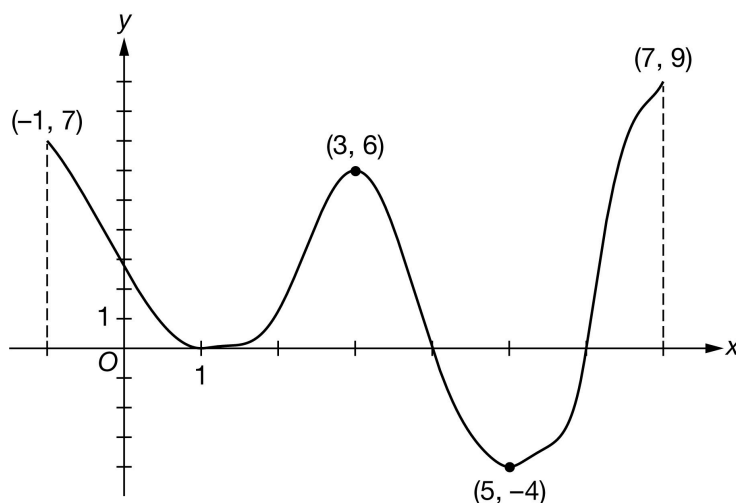
5-5

19. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The figure above shows the graph of h' , the derivative of a twice-differentiable function h , on the closed interval $-1 \leq x \leq 7$. The graph of h' has horizontal tangent lines at $x = 1$, $x = 3$, and $x = 5$. The areas of the regions bounded by the x -axis and the graph of h' on the intervals $[-1, 1]$, $[1, 4]$, $[4, 6]$, and $[6, 7]$ are 6, 8, 5.6, and 6, respectively. The function h is defined for all real numbers and satisfies $h(5) = 5$ and $h(7) = 8$.

- Find the x -coordinate of each critical point of h on the open interval $-1 < x < 7$.
- Identify the critical point from part (a) which corresponds to a relative minimum of h . Give a reason for your answer.
- Find the value of h at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of h on $-1 \leq x \leq 7$. Justify your answer.
- On what open intervals is h both increasing and concave up? Give a reason for your answer.
- Explain why there must be a value of p , for $5 < p < 7$, such that $h(p) = 6.5$.
- Find $\lim_{x \rightarrow 7^-} \frac{h(x) - 8}{3x - 21}$. Give a reason for your answer.
- The function a is defined by $a(x) = (h(x))^2$. Find $a'(5)$. Show the computations that lead to your answer.

Part A

5-5

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = 1$, $x = 4$, and $x = 6$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1, 7)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1, 7)$. The point is still earned provided the three critical points, $x = 1$, $x = 4$, and $x = 6$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(1, 0)$, $(4, 0)$, $(6, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = 0$ at $x = 1$, $x = 4$, and $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 6 and give the correct reason of “ h' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ h switches from decreasing to increasing at $x = 6$.”
- Saying that “ $h'(6)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of h' , the one given in the stem of the problem.

A response indicating that “ $h'(x)$ is negative for $x < 6$ and positive for $x > 6$ ” is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.

5-5



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative minimum value of h occurs where $h'(x)$ changes from negative to positive values.

$x = 6$ is the only location on $-1 < x < 7$ for which this is true, so $x = 6$ is the only location of a relative minimum value of h on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -1$, $x = 1$, or $x = 4$ is selected as the location of the local minimum in part (b). Note that $h(-1) = -6.4$, $h(1) = -0.4$, and $h(4) = 7.6$. No other x -coordinates will be read for consistency.

The response $h(6) = h(7) - 6 = 2$ earns both points.

The responses of both $8 - 6 = 2$ and $h(6) = 2$ do not earn the first point, but earn the second point, provided no other work is presented.

The string of equalities $h(6) = h(7) - \int_6^7 h'(x) dx = 8 - 6 = 2$ earns both points.

Special Case: If in part (b), a response picks $x = 5$ or $x = 7$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.

5-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $h(6) = h(7) + \int_7^6 h'(x) dx$

☐ Answer

Solution:

$$\begin{aligned} h(6) &= h(7) + \int_7^6 h'(x) dx \\ &= h(7) - \int_6^7 h'(x) dx = 8 - 6 = 2 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 6$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 5$), and thus the third point cannot be earned.

It is possible to not refer to $h(7) = 8$ by indicating that h' is positive on the interval $(6, 7)$, thus eliminating 8 as a possible absolute minimum.

A complete list of given or computable values for $h(x)$ is given in the table below

5-5

x	$h(x)$
-1	-6.4
1	-0.4
4	7.6
5	5
6	2
7	8



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 6$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$h(-1) = h(7) + \int_7^{-1} h'(x) dx$$

$$= h(7) - \int_{-1}^7 h'(x) dx = 8 - (6 + 8 - 5.6 + 6) = -6.4$$

$$h(6) = 2$$

$$h(7) = 8$$

Therefore, the absolute minimum value of h on $-1 \leq x \leq 7$ is $h(-1) = -6.4$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, for example “(1, 3] and [6, 7)” earns the first point.

The reason “ h' is increasing” can be expressed by either “ h'' is positive” or “the slope of h' is positive.”

We will overlook the false implication that $h' > 0$ on $[1, 3]$ and $[6, 7]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[1, 3] \cup [6, 7]$.

5-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where h' is positive.

h is concave up where h' is increasing.

h is both increasing and concave up on the open intervals $(1, 3)$ and $(6, 7)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that h is continuous because h is differentiable or h is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[5, 7]$, or the point is not earned.

The y -values of h need to be mentioned by either referring to the numerical values of 5 and 8, or mentioning $h(5)$ and $h(7)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since h is differentiable it’s continuous and 6.5 is between 5 and 8 such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

5-5

h is differentiable. $\Rightarrow h$ is continuous.

$$h(5) = 5 < 6.5 < 8 = h(7)$$

By the Intermediate Value Theorem, there must be a value of p , for $5 < p < 7$, such that $h(p) = 6.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of h' or h .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $h(7) - 8 = 0$ and $3 \cdot 7 - 21 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $h(7) = 8$ and $3 \cdot 7 = 21$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h(7)-8}{3 \cdot 7-21} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h'(x)}{3} = \frac{9}{3}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{0}{0}$ and

$\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h'(x)}{3}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of 3, or its numeric equivalent.

5-5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h is twice differentiable. $\Rightarrow$ both h' and h are continuous. $\Rightarrow \lim_{x \rightarrow 7^-} h(x) = h(7)$ and $\lim_{x \rightarrow 7^-} h'(x) = h'(7)$.

$$\lim_{x \rightarrow 7^-} (h(x) - 8) = h(7) - 8 = 8 - 8 = 0$$

$$\lim_{x \rightarrow 7^-} (3x - 21) = 3 \cdot 7 - 21 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 7^-} \frac{h(x) - 8}{3x - 21} &= \lim_{x \rightarrow 7^-} \frac{h'(x)}{3} \\ &= \frac{h'(7)}{3} = \frac{9}{3} = 3 \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The Power rule point can be earned by just seeing $2 \cdot h(x)$ or $2 \cdot h(5)$.

The chain rule point can be earned by seeing an expression involving h but not h' times $h'(5)$.

All three points will be awarded for seeing only $2 \cdot 5 \cdot (-4)$, but $10 \cdot (-4)$, $2 \cdot (-20)$, or $-8 \cdot 5$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

5-5

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$a'(x) = 2h(x) \cdot h'(x)$$

$$a'(5) = 2 \cdot h(5) \cdot h'(5) = 2 \cdot 5 \cdot (-4) = -40$$

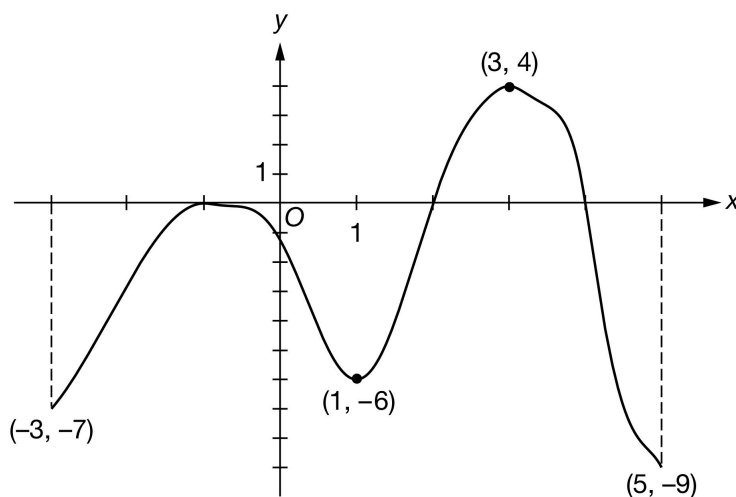
5-5

20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the closed interval $-3 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of f' on the intervals $[-3, -1]$, $[-1, 2]$, $[2, 4]$, and $[4, 5]$ are 6, 8, 5.6, and 6, respectively. The function f is defined for all real numbers and satisfies $f(3) = 10$ and $f(5) = 7$.

- Find the x -coordinate of each critical point of f on the open interval $-3 < x < 5$.
- Identify the critical point from part (a) which corresponds to a relative minimum of f . Give a reason for your answer.
- Find the value of f at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of f on $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals is f both decreasing and concave down? Give a reason for your answer.
- Explain why there must be a value of d , for $3 < d < 5$, such that $f(d) = 8.5$.
- Find $\lim_{x \rightarrow 5^-} \frac{2f(x) - 14}{x^2 - 25}$. Give a reason for your answer.
- The function m is defined by $m(x) = (f(x))^2$. Find $m'(3)$. Show the computations that lead to your answer.

Part A

5-5

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -1$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(2, 0)$, $(4, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$f'(x) = 0$ at $x = -1$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 2 and give the correct reason of “ f' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ f switches from decreasing to increasing at $x = 2$ ”
- Saying that “ $f'(2)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of f' , the one given in the stem of the problem.

A response indicating that “ $f'(x)$ is negative for $x < 2$ and positive for $x > 2$ ” is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.

5-5



0

1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative minimum value of f occurs where $f'(x)$ changes from negative to positive values.

$x = 2$ is the only location on $-3 < x < 5$ for which this is true, so $x = 2$ is the only location of a relative minimum value of f on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = -1$, or $x = 4$ is selected as the location of the local minimum in part (b). Note that $f(-3) = 21.4$, $f(-1) = 15.4$, and $f(4) = 13$. No other x -coordinates will be read for consistency.

The response $f(2) = f(5) - (5.6 - 6) = 7.4$ earns both points.

The responses of both $7 - (-0.4) = 7.4$ and $f(2) = 7.4$ do not earn the first point, but earn the second point, provided no other work is presented.

The string of equalities $f(2) = f(5) - \int_2^5 f'(x) dx = 7 - (-0.4) = 7.4$ earns both points.

Special Case: If in part (b), a response picks $x = 3$ or $x = 5$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.

5-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $f(2) = f(5) + \int_5^2 f'(x) dx$

☐ Answer

Solution:

$$\begin{aligned} f(2) &= f(5) + \int_5^2 f'(x) dx = f(5) - \int_2^5 f'(x) dx \\ &= f(5) - \int_2^4 f'(x) dx - \int_4^5 f'(x) dx = 7 - 5.6 - (-6) = 7.4 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 2$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 3$), and thus the third point cannot be earned.

It is possible to not refer to $f(-3) = 21.4$ by indicating that f' is negative on the interval $(-3, -1)$, thus eliminating 21.4 as a possible absolute minimum.

A complete list of given or computable values for $f(x)$ is given in the table below.

5-5

x	$f(x)$
-3	21.4
-1	15.4
2	7.4
3	10
4	13
5	7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$f(-3) = f(5) + \int_5^{-3} f'(x) dx$$

$$= f(5) - \int_{-3}^5 f'(x) dx = 7 - (-6 + 5.6 - 8 - 6) = 21.4$$

$$f(2) = 7.4$$

$$f(5) = 7$$

So the absolute minimum value of f on $-3 \leq x \leq 5$ is

$$f(5) = 7.$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, for example “ $(-1, 1]$ and $[4, 5)$ ” earns the first point.

The reason “ f' is decreasing” can be expressed by either “ f'' is negative” or “the slope of f' is negative.”

We will overlook the false implication that $f' < 0$ on $[-1, 1]$ and $[4, 5]$.

5-5

The second point can be earned without earning the first point only if the given answer is a subset of $[-1, 1] \cup [4, 5]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

f is decreasing where f' is negative.

f is concave down where f' is decreasing.

f is both decreasing and concave down on the open intervals $(-1, 1)$ and $(4, 5)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that f is continuous because f is differentiable or f is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[3, 5]$, or the point is not earned.

The y -values of f need to be mentioned by either referring to the numerical values of 7 and 10, or mentioning $f(3)$ and $f(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since f is differentiable it’s continuous and 8.5 is between 7 and 10 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

5-5

Solution:

f is differentiable. $\Rightarrow f$ is continuous.

$$f(5) = 7 < 8.5 < 10 = f(3)$$

By the Intermediate Value Theorem, there must be a value of d , for $3 < d < 5$, such that $f(d) = 8.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of f' or f .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $2f(5) - 14 = 0$ and $5^2 - 25 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $2f(5) = 14$ and $5^2 = 25$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f(5)-14}{5^2-25} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f'(x)}{2x} = \frac{-9}{5}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{0}{0}$ and

$\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

5-5

The third point is earned only for the answer of $-\frac{9}{5}$, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

f is twice differentiable. $\Rightarrow$ both f' and f are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} f(x) = f(5)$ and $\lim_{x \rightarrow 5^-} f'(x) = f'(5)$.

$$\lim_{x \rightarrow 5^-} (2f(x) - 14) = 2 \cdot f(5) - 14 = 14 - 14 = 0$$

$$\lim_{x \rightarrow 5^-} (x^2 - 25) = 5^2 - 25 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{2f(x) - 14}{x^2 - 25} &= \lim_{x \rightarrow 5^-} \frac{2f'(x)}{2x} \\ &= \frac{2f'(5)}{2 \cdot 5} = \frac{-9}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $2 \cdot f(x)$ or $2 \cdot f(3)$.

The chain rule point can be earned by seeing an expression involving f but not f' times $f'(3)$.

All three points will be awarded for seeing only $2 \cdot 10 \cdot 4$, but $20 \cdot 4$, $2 \cdot 40$, or $8 \cdot 10$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

5-5

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$m'(x) = 2f(x) \cdot f'(x)$$

$$m'(3) = 2 \cdot f(3) \cdot f'(3) = 2 \cdot 10 \cdot 4 = 80$$

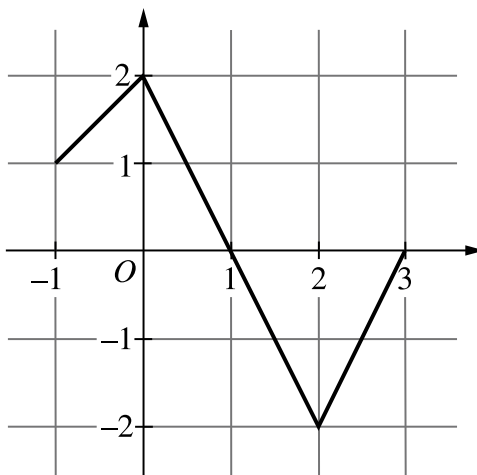
5-5

21. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

Let f be the continuous function defined on the closed interval $[-1, 3]$ whose graph, consisting of three line segments, is shown. Let g be the function given by $g(x) = \int_0^x f(t) \, dt$.

- On what intervals, if any, is g decreasing? Give a reason for your answer.
- Find $g''(-\frac{1}{2})$, or explain why it does not exist.
- What is the absolute minimum value of g on the closed interval $[-1, 3]$? Justify your answer.
- Let h be the function defined by $h(x) = \ln(g(x))$ for $0 < x < 2$. Find $h'(\frac{3}{2})$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $g'(x) = f(x)$ in any part of the problem.

• A response of $g''(x) = f'(x)$ or $g''(-\frac{1}{2}) = f'(-\frac{1}{2})$ in part (b) is sufficient to earn the first point.

The interval $[1, 3]$ can be reported $(1, 3)$, $(1, 3]$, or $[1, 3]$.

5-5

The second point can be earned without the first by presenting a correct interval and a statement that $f(x) < 0$ on this interval.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $g'(x) = f(x)$ in any part of response
- ☐ Answer with reason

Solution:

$g'(x) = f(x) < 0$ on the open interval $(1, 3)$. Therefore, g is decreasing on the closed interval $[1, 3]$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An answer of 1 with no supporting work earns the point.

Any work shown must be correct in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$g''(x) = f'(x)$$

$$g''\left(-\frac{1}{2}\right) = f'\left(-\frac{1}{2}\right) = 1$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for any presented consideration of $x = -1$ and $x = 3$.

A response that finds only that $g(-1) = -1.5$ and declares this as the minimum earns only the second point.

5-5

A response that evaluates $g(x)$ at any values of x in addition to $x = -1$, $x = 1$, and $x = 3$ does not earn the third point.

To earn the third point, a response must:

- have earned the first point,
- have no errors in the evaluation of $g(-1)$ and $g(3)$,
- conclude that $g(3) > g(-1)$, and
- either evaluate $g(1)$ as part of the Candidates Test or include a global argument that specifically addresses the behavior of $g(x)$ at $x = 1$ (e.g., “ $g'(x) = f(x)$ changes signs from positive to negative at $x = 1$ ” or “ $g'(1) = f(1) = 0$ and there is a maximum at $x = 1$ ”).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = -1$ and $x = 3$
- ☐ Answer
- ☐ Justification

Solution:

Because $g'(x) = f(x) > 0$ for $-1 < x < 1$ and $g'(x) = f(x) < 0$ for $1 < x < 3$, the minimum value of g must occur at either $x = -1$ or $x = 3$.

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -1.5$$

$$g(3) = \int_0^3 f(t) dt = -1$$

Therefore, the minimum value of g is $g(-1) = -1.5$.

Alternate solution using Candidates Test:

$$g'(x) = f(x) = 0 \Rightarrow x = 1$$

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -1.5$$

$$g(1) = \int_0^1 f(t) dt = 1$$

5-5

$$g(3) = \int_0^3 f(t) dt = -1$$

Therefore, the minimum value of g is $g(-1) = -1.5$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $h'(x) = \frac{1}{g(x)} \cdot g'(x)$, $h'\left(\frac{3}{2}\right) = \frac{1}{g\left(\frac{3}{2}\right)} \cdot g'\left(\frac{3}{2}\right)$, or $\frac{1}{\frac{3}{4}} \cdot (-1)$.

To earn the second point a response must use the correct value of either $g\left(\frac{3}{2}\right)$ or $g'\left(\frac{3}{2}\right)$ (or both) in an expression for $h'\left(\frac{3}{2}\right)$.

The minimal answer that earns all 3 points is $\frac{1}{\frac{3}{4}} \cdot (-1)$.

An answer of $-\frac{4}{3}$ with incorrect or without supporting work does not earn any points.

The third point is earned only for $-\frac{4}{3}$ or equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $h'(x)$
- ☐ Correct value of $g\left(\frac{3}{2}\right)$ or $g'\left(\frac{3}{2}\right)$
- ☐ Answer

Solution:

$$h'(x) = \frac{1}{g(x)} \cdot g'(x)$$

$$h'\left(\frac{3}{2}\right) = \frac{1}{g\left(\frac{3}{2}\right)} \cdot g'\left(\frac{3}{2}\right) = \frac{1}{\frac{3}{4}} \cdot (-1) = -\frac{4}{3}$$

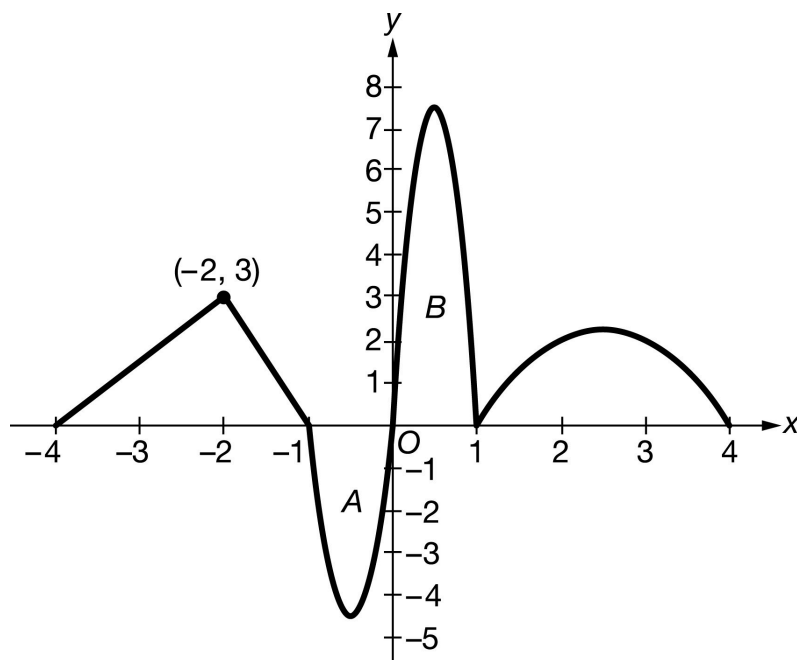
5-5

22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f is defined for $-4 \leq x \leq 4$. The graph of f , shown above, consists of two line segments and portions of three parabolas. The graph has horizontal tangents at $x = -\frac{1}{2}$, $x = \frac{1}{2}$, and $x = \frac{5}{2}$. It is known that $f(x) = -x^2 + 5x - 4$ for $1 \leq x \leq 4$. The areas of regions A and B bounded by the graph of f and the x -axis are 3 and 5, respectively. Let g be the function defined by

$$g(x) = \int_{-4}^x f(t) \, dt.$$

- Find $g(0)$ and $g(4)$.
- Find the absolute minimum value of g on the closed interval $[-4, 4]$. Justify your answer.
- Find all intervals on which the graph of g is concave down. Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Simplification is not required for the first point.

5-5

The second point is earned with a correct integral or with work that shows that the integral has been taken. For

example, $\int_1^4 f(t) dt$, or $\int_1^4 (-t^2 + 5t - 4) dt$, or $[-\frac{1}{3}t^3 + \frac{5}{2}t^2 - 4t]_1^4$ would earn the second point.

The third point is earned with $-\frac{1}{3}t^3 + \frac{5}{2}t^2 - 4t$.

Simplification is not required for the fourth point.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

- ☐ $g(0)$
- ☐ integral of f over $1 \leq t \leq 4$
- ☐ antiderivative
- ☐ $g(4)$

Solution:

$$g(0) = \int_{-4}^0 f(t) dt = \frac{9}{2} - 3 = \frac{3}{2}$$

$$\begin{aligned} g(4) &= \int_{-4}^4 f(t) dt \\ &= \int_{-4}^0 f(t) dt + \int_0^1 f(t) dt + \int_1^4 f(t) dt \\ &= \frac{3}{2} + 5 + \int_1^4 (-t^2 + 5t - 4) dt \\ &= \frac{3}{2} + 5 + [-\frac{1}{3}t^3 + \frac{5}{2}t^2 - 4t]_1^4 \\ &= \frac{3}{2} + 5 + [(-\frac{1}{3} \cdot 4^3 + \frac{5}{2} \cdot 4^2 - 4 \cdot 4) - (-\frac{1}{3} \cdot 1^3 + \frac{5}{2} \cdot 1^2 - 4 \cdot 1)] \\ &= \frac{3}{2} + 5 + (\frac{8}{3} - (-\frac{11}{6})) = 11 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an explicit statement connecting $g' = f$. Note that it is possible that this connection is made in an earlier part and will earn the point here.

The second point is earned for values appearing in an equation, table, or paragraph.

5-5

The third point is earned by ruling out all other candidates and committing to the minimum value of 0.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $g'(x) = f(x)$
- ☐ identifies $x = -4$ and $x = 0$ as candidates
- ☐ answer with justification

Solution:

$g'(x) = f(x)$ is negative for $-1 < x < 0$, and nonnegative elsewhere. Thus, the absolute minimum value of g on $[-4, 4]$ can only occur at $x = -4$ or $x = 0$.

$$g(-4) = 0$$

$$g(0) = \frac{3}{2}$$

The absolute minimum value of g on $[-4, 4]$ is $g(-4) = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with all three intervals, with or without the endpoints. Verbal expressions rather than intervals also earn the point.

The second point is earned with reasoning clearly tied to the given graph of f .



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ intervals
- ☐ reason

Solution:

The graph of g is concave down on the intervals $-2 < x < -\frac{1}{2}$, $\frac{1}{2} < x < 1$, and $\frac{5}{2} < x < 4$ because $g'(x) = f(x)$ is decreasing on these intervals.

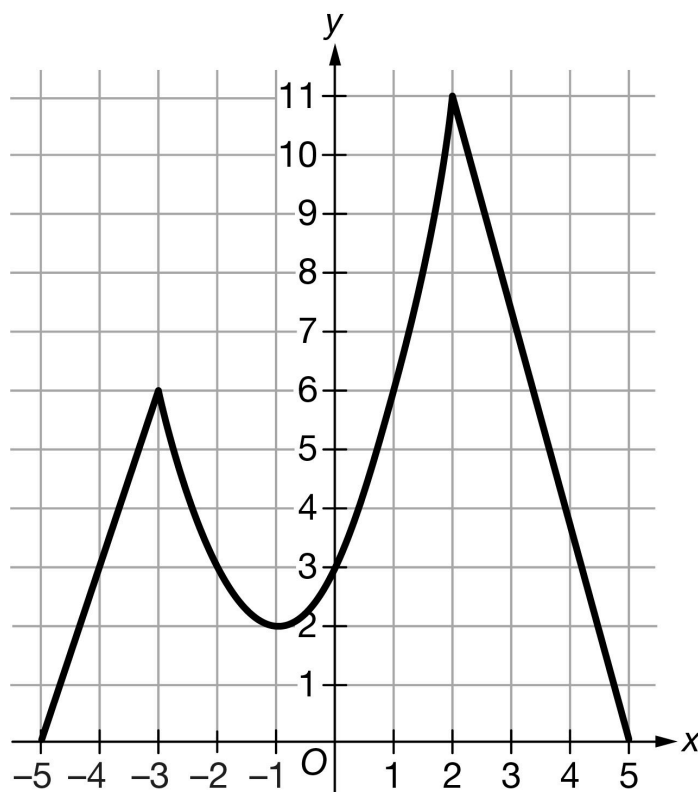
5-5

23. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f is defined on the closed interval $[-5, 5]$. The graph of f consists of a parabola and two line segments, as shown in the figure above. Let g be a function such that $g'(x) = f(x)$.

(a) Fill in the missing entries in the table below to describe the behavior of f' and f'' . Indicate Positive, Negative, or 0. Give reasons for your answers.

x	$-5 < x < -3$	$-3 < x < -1$	$-1 < x < 2$	$2 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$				
$f''(x)$				

(b) There is no value of x in the open interval $(-1, 5)$ at which $f'(x) = \frac{f(5) - f(-1)}{5 - (-1)}$. Explain why this does not violate the Mean Value Theorem.

5-5

- (c) Find all values of x in the open interval $(-5, 5)$ at which the graph of g has a point of inflection. Explain your reasoning.
- (d) At what value of x does g attain its absolute maximum on the closed interval $[-5, 5]$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ behavior for $f'(x)$
- ☐ behavior for $f''(x)$
- ☐ reasons

Solution:

x	$-5 < x < -3$	$-3 < x < -1$	$-1 < x < 2$	$2 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$	Positive	Negative	Positive	Negative
$f''(x)$	0	Positive	Positive	0

$f'(x)$ is positive for $-5 < x < -3$ and $-1 < x < 2$ because f is increasing there.

$f'(x)$ is negative for $-3 < x < -1$ and $2 < x < 5$ because f is decreasing there.

$f''(x)$ is positive for $-3 < x < -1$ and $-1 < x < 2$ because the graph of f is concave up there.

$f''(x) = 0$ for $-5 < x < -3$ and $2 < x < 5$ because the graph of f is linear there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct explanation.

Solution:

5-5

This does not violate the Mean Value Theorem because f is not differentiable at $x = 2$ due to a sharp corner. The Mean Value Theorem cannot be applied on the interval $-1 < x < 5$.

Part C

The response is not eligible for any points if incorrect values of x are included.

A response may earn the first point for two correct values of x without explanations or with incomplete, incorrect explanations.

A maximum of 1 point may be earned for three correct values of x without explanations.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ One correct value of x with explanation OR Two correct values of x without explanations
- ☐ A second correct value of x with explanation
- ☐ A third correct value of x with explanation

Solution:

The graph of g has a point of inflection at $x = -3$, $x = -1$, and $x = 2$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -3$ and $x = 2$, and $g'(x) = f(x)$ changes from decreasing to increasing at $x = -1$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ reason

Solution:

5-5

Because $g'(x) = f(x) > 0$ on the interval $(-5, 5)$, g is increasing on the interval $(-5, 5)$. Therefore, the absolute maximum value of g on the closed interval $[-5, 5]$ occurs at the right endpoint $x = 5$.

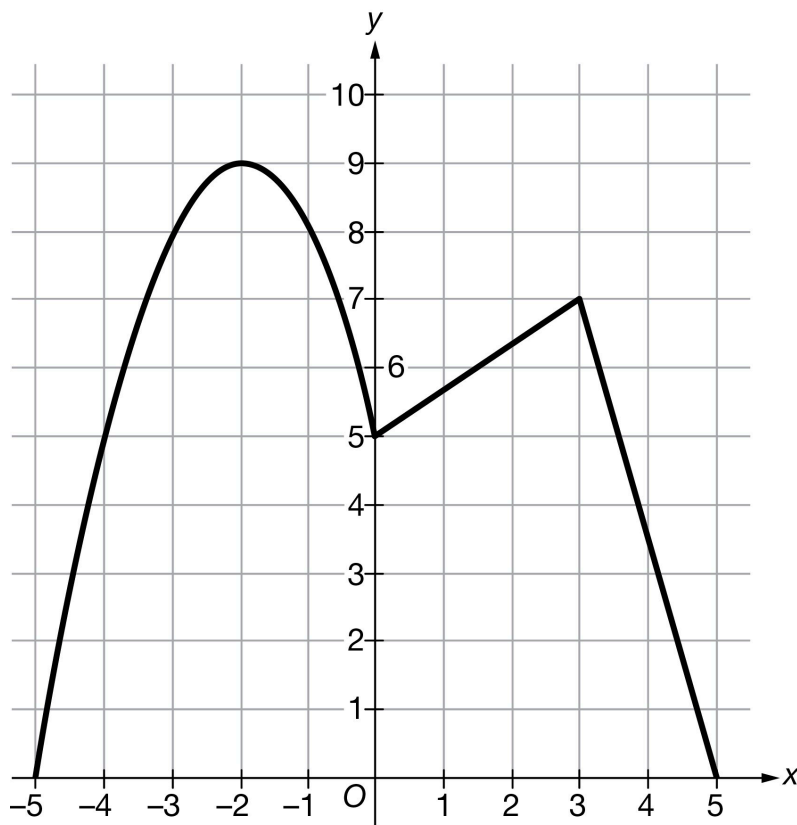
5-5

24. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f is defined on the closed interval $[-5, 5]$. The graph of f consists of a parabola and two line segments, as shown in the figure above. Let g be a function such that $g'(x) = f(x)$.

(a) Fill in the missing entries in the table below to describe the behavior of f' and f'' . Indicate Positive, Negative, or 0. Give reasons for your answers.

x	$-5 < x < -2$	$-2 < x < 0$	$0 < x < 3$	$3 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$				
$f''(x)$				

5-5

- (b) There is no value of x in the open interval $(-1, 3)$ at which $f'(x) = \frac{f(3)-f(-1)}{3-(-1)}$. Explain why this does not violate the Mean Value Theorem.
- (c) Find all values of x in the open interval $(-5, 5)$ at which the graph of g has a point of inflection. Explain your reasoning.
- (d) At what value of x does g attain its absolute maximum on the closed interval $[-5, 5]$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ behavior for $f'(x)$
- ☐ behavior for $f''(x)$
- ☐ reasons

Solution:

x	$-5 < x < -2$	$-2 < x < 0$	$0 < x < 3$	$3 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$	Positive	Negative	Positive	Negative
$f''(x)$	Negative	Negative	0	0

$f'(x)$ is positive for $-5 < x < -2$ and $0 < x < 3$ because f is increasing there.

$f'(x)$ is negative for $-2 < x < 0$ and $3 < x < 5$ because f is decreasing there.

$f''(x)$ is negative for $-5 < x < -2$ and $-2 < x < 0$ because the graph of f is concave down there.

$f''(x) = 0$ for $0 < x < 3$ and $3 < x < 5$ because the graph of f is linear there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

5-5

The student response accurately includes a correct explanation.

Solution:

This does not violate the Mean Value Theorem because f is not differentiable at $x = 0$ due to a sharp corner. The Mean Value Theorem cannot be applied on the interval $-1 < x < 3$.

Part C

Two values of x must be correct without explanation to earn one point, and no incorrect values included.

A maximum of 1 point may be earned for one correct value with explanation.

A maximum of 2 points may be earned for two correct values with explanation.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ values of x (2 points full credit)
- ☐ explanation

Solution:

The graph of g has a point of inflection at $x = -2$, $x = 0$, and $x = 3$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -2$ and $x = 3$, and $g'(x) = f(x)$ changes from decreasing to increasing at $x = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ reason

5-5**Solution:**

Because $g'(x) = f(x) > 0$ on the interval $(-5, 5)$, g is increasing on the interval $(-5, 5)$. Therefore, the absolute maximum value of g on the closed interval $[-5, 5]$ occurs at the right endpoint $x = 5$.

5-5

25.  A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \leq t \leq 12$. At time $t = 0$, when the sale begins, there are no shoppers in the store.

(a) At what rate are shoppers entering the store 3 hours after the start of the sale?

(b) Find the value of $\frac{1}{3} \int_6^9 S'(t) dt$. Using correct units, explain the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$ in the context of this problem.

(c) The rate at which shoppers leave the store, measured in shoppers per hour, is modeled by the function L defined by $L(t) = -80 + \frac{4400}{t^2 - 14t + 55}$ for $0 \leq t \leq 12$. According to the model, how many shoppers are in the store at the end of the sale (time $t = 12$)? Give your answer to the nearest whole number.

(d) Using the given models, find the time t , $0 \leq t \leq 12$, at which the number of shoppers in the store is the greatest. Justify your answer.

Part A

The response can earn up to 1 point:

1 point is earned for the answer

$$S'(3) = 486 \text{ shoppers/hour}$$



0	1
---	---

The student earns all of the following point:

1 point is earned for the answer

$$S'(3) = 486 \text{ shoppers/hour}$$

Part B

The response can earn up to 2 points:

1 point is earned for the value of $\frac{1}{3} \int_6^9 S'(t) dt$

$$\begin{aligned} \frac{1}{3} \int_6^9 S'(t) dt &= \frac{1}{3}(S(9) - S(6)) \\ &= \frac{1}{3}(3280.5 - 2376) = 301.5 \end{aligned}$$

1 point is earned for the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$

5-5

$\frac{1}{3} \int_6^9 S'(t) dt$ is the average rate at which shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the value of $\frac{1}{3} \int_6^9 S'(t) dt$

$$\begin{aligned} \frac{1}{3} \int_6^9 S'(t) dt &= \frac{1}{3}(S(9) - S(6)) \\ &= \frac{1}{3}(3280.5 - 2376) = 301.5 \end{aligned}$$

1 point is earned for the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$

$\frac{1}{3} \int_6^9 S'(t) dt$ is the average rate at which shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.

Part C

The response can earn up to 3 points:

1 point is earned for the integral

1 point is for use of $S(12)$

1 point is for the answer

$$S(12) - \int_0^{12} L(t) dt = 195.701684$$

Therefore, there are approximately 196 shoppers in the store at the end of the sale.



0	1	2	3
---	---	---	---

The student earns all of the following points:

1 point is earned for the integral

5-5

1 point is for use of $S(12)$

1 point is for the answer

$$S(12) - \int_0^{12} L(t) dt = 195.701684$$

Therefore, there are approximately 196 shoppers in the store at the end of the sale.

Part D

The response can earn up to 3 points:

1 point is earned for considering $S'(t) - L(t) = 0$

1 point is earned for the answer

1 point is earned for justification

The number of shoppers in the store at time t is given by the function N defined by

$$N(t) = S(t) - \int_0^t L(x) dx$$

$$N'(t) = S'(t) - L(t) = 0 \text{ when } t = 5.545066$$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.545$ hours.



0	1	2	3
---	---	---	---

The student earns all of the following points:

1 point is earned for considering $S'(t) - L(t) = 0$

1 point is earned for the answer

1 point is earned for justification

The number of shoppers in the store at time t is given by the function N defined by

5-5

$$N(t) = S(t) - \int_0^t L(x) \, dx$$

$$N'(t) = S'(t) - L(t) = 0 \text{ when } t = 5.545066$$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.545$ hours.

5-5

26. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The number of mosquitoes in a field after a major rainfall is modeled by the function M defined by $M(t) = -t^3 + 12t^2 + 144t$, where t is the number of days after the rainfall ended and $0 \leq t \leq 18$.

- Using correct units, interpret the meaning of $M'(15) = -171$ in the context of the problem.
- Based on the model, what is the absolute maximum number of mosquitoes in the field over the time interval $0 \leq t \leq 18$? Justify your answer.
- For what values of t is the rate of change of the number of mosquitoes in the field increasing?
- For $0 \leq t \leq 18$, the number of bats in the field is modeled by the differentiable function B , where B is a function of the number of mosquitoes in the field. Based on the models, write an expression for the rate of change of the number of bats in the field at time $t = a$.

Part A

The point requires a time reference, rate, and units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct interpretation of $M'(15) = -171$

Solution:

At time $t = 15$ days, the number of mosquitoes in the field is decreasing at a rate of 171 mosquitoes per day.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $M'(t)$ with a maximum of one computational error.

5-5

The fourth point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $M'(t)$
- ☐ sets $M'(t) = 0$
- ☐ identifies $t = 12$ as a candidate
- ☐ answer with justification

Solution:

$$M'(t) = -3t^2 + 24t + 144 = -3(t^2 - 8t - 48) = -3(t - 12)(t + 4)$$

$$M'(t) = 0 \Rightarrow t = 12$$

Because $M'(t)$ changes from positive to negative at $t = 12$, M has a relative maximum at $t = 12$.

Because $t = 12$ is the only critical point of M on the interval $0 \leq t \leq 18$, that is the location of a relative maximum. It is also the location of the absolute maximum of M on the interval $0 \leq t \leq 18$.

$$M(12) = -12^3 + 12(12^2) + 144(12) = 144(12) = 1728$$

The absolute maximum number of mosquitoes in the field over the time interval $0 \leq t \leq 18$ is $M(12) = 1728$.

Part C

The third point may be earned based on a consistent answer using an incorrect $M''(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ considers $M''(t)$
- ☐ expression for $M''(t)$

5-5

☐ answer
Solution:

$$M''(t) = -6t + 24 = -6(t - 4)$$

$$M''(t) = -6(t - 4) > 0 \Rightarrow t < 4$$

The rate of change of the number of mosquitoes in the field is increasing for $0 \leq t \leq 4$.

Part D

The point is earned with $B'(M(a)) \cdot M'(a)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct expression.

Solution:

The number of bats in the field at time $t = a$ is given by $B(M(a))$.

Therefore, the rate of change of the number of bats in the field at time $t = a$ is given by

$$B'(M(a)) \cdot M'(a) = B'(-a^3 + 12a^2 + 144a) \cdot (-3a^2 + 24a + 144).$$

5-5

27. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The number of fish in a small bay is modeled by the function F defined by $F(t) = 10(t^3 - 12t^2 + 45t + 100)$, where t is measured in days and $0 \leq t \leq 8$.

- Using correct units, interpret the meaning of $F'(4) = -30$ in the context of the problem.
- Based on the model, what is the absolute minimum number of fish in the bay over the time interval $0 \leq t \leq 8$? Justify your answer.
- For what values of t , $0 \leq t \leq 8$, is the rate of change of the number of fish in the bay decreasing?
- For $0 \leq t \leq 8$, the number of pelicans flying near the bay is modeled by the differentiable function P , where P is a function of the number of fish in the bay. Based on the models, write an expression for the rate of change of the number of pelicans flying near the bay at time $t = c$.

Part A

The point requires a time reference, rate, and units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1
---	---

The student response accurately includes a correct interpretation of $F'(4) = -30$.

Solution:

At time $t = 4$ days, the number of fish in the bay is decreasing at a rate of 30 fish per day.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $F'(t)$ with a maximum of one computational error.

5-5

The fourth point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $F'(t)$
- ☐ sets $F'(t) = 0$
- ☐ identifies $t = 5$ as a candidate
- ☐ answer with justification

Solution:

$$F'(t) = 10(3t^2 - 24t + 45) = 10 \cdot 3(t^2 - 8t + 15) = 30(t - 3)(t - 5)$$

$$F'(t) = 0 \Rightarrow t = 3 \text{ and } t = 5$$

Because $F'(t)$ changes from positive to negative at $t = 3$ and from negative to positive at $t = 5$, consider $t = 0$ and $t = 5$ for the location of the absolute minimum. F has a relative minimum at $t = 5$.

$$F(0) = 1000$$

$$F(5) = 10(5^3 - 12(5)^2 + 45(5) + 100) = 1500$$

The absolute minimum number of fish in the bay over the time interval

$$0 \leq t \leq 8 \text{ is } F(0) = 1000.$$

Part C

The third point may be earned based on a consistent answer using an incorrect $F''(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

5-5

- ☐ considers $F''(t)$
- ☐ expression for $F''(t)$
- ☐ answer

Solution:

$$F''(t) = 10(6t - 24) = 60(t - 4)$$

$$F''(t) = 60(t - 4) < 0 \Rightarrow t < 4$$

The rate of change of the number of fish in the bay is decreasing for $0 \leq t \leq 4$.

Part D

The point is earned with $P'(F(c)) \cdot F'(c)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct expression.

Solution:

The number of pelicans flying near the bay at time $t = c$ is given by $P(F(c))$.

Therefore, the rate of change of the number of pelicans flying near the bay at time $t = c$ is given by

$$P'(F(c)) \cdot F'(c) = P'(10(c^3 - 12c^2 + 45c + 100)) \cdot (10(3c^2 - 24c + 45)).$$

5-5

28. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	5	15	30	35
$A(t)$ (gallons)	10	18	25	16	8

The number of gallons of olive oil in a tank at time t is given by the twice-differentiable function A , where t is measured in hours and $0 \leq t \leq 35$. Values of $A(t)$ at selected times t are given in the table above.

- (a) Use the data in the table to estimate the rate at which the number of gallons of olive oil in the tank is changing at time $t = 10$ hours. Show the computations that lead to your answer. Indicate units of measure.
- (b) For $0 \leq t \leq 30$, is there a time t at which $A'(t) = \frac{1}{5}$? Justify your answer.
- (c) The number of gallons of olive oil in the tank at time t is also modeled by the function G defined by $G(t) = 5t - \frac{2}{3}(t+9)^{\frac{3}{2}} + 28$, where t is measured in hours and $0 \leq t \leq 35$. Based on the model, at what time t , for $0 \leq t \leq 35$, is the number of gallons of olive oil in the tank an absolute maximum? Justify your answer.
- (d) For the function G defined in part (c), the locally linear approximation near $t = 7$ is used to approximate $G(8)$. Is this approximation an overestimate or an underestimate for the value of $G(8)$? Give a reason for your answer.

Part A

Substitution of function values is required. The answer does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ approximation with units

Solution:

5-5

$$A'(10) \approx \frac{A(15) - A(5)}{15 - 5} = \frac{25 - 18}{15 - 5} = \frac{7}{5} \text{ gallons per hour}$$

Part B

The first point requires identification of the difference quotient. The substitution of function values and conclusion impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{A(30) - A(0)}{30 - 0}$
- ☐ justification using Mean Value Theorem

Solution:

A is twice differentiable. $\Rightarrow A$ is differentiable. $\Rightarrow A$ is continuous.

$$\frac{A(30) - A(0)}{30 - 0} = \frac{16 - 10}{30 - 0} = \frac{6}{30} = \frac{1}{5}$$

Therefore, by the Mean Value Theorem, there is a time t , for $0 \leq t \leq 30$, at which $A'(t) = \frac{1}{5}$.

Part C

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $G'(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $G'(t)$

5-5

- ☐ sets $G'(t) = 0$
- ☐ identifies $t = 16$ as a candidate
- ☐ answer with justification

Solution:

$$G'(t) = 5 - (t + 9)^{\frac{1}{2}} = 0 \Rightarrow (t + 9)^{\frac{1}{2}} = 5$$

$$\Rightarrow t + 9 = 25 \Rightarrow t = 16$$

Because $G'(t)$ changes from positive to negative at $t = 16$, G has a relative maximum at $t = 16$.

Since $t = 16$ is the only critical point of G on the interval $0 \leq t \leq 35$ that is the location of a relative maximum, it is also the location of the absolute maximum of G on the interval $0 \leq t \leq 35$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2
---	---	---



The student response accurately includes both of the criteria below.

- ☐ $G''(t)$
- ☐ overestimate with reason

Solution:

$$G''(t) = -\frac{1}{2}(t + 9)^{-\frac{1}{2}}$$

$G''(t) < 0$ for $7 \leq t \leq 8$, and so the graph of G is concave down for $7 \leq t \leq 8$. Therefore, the approximation of $G(8)$ using the locally linear approximation of G near $t = 7$ is an overestimate.

5-5

29. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	5	8	10
$G(t)$ (thousands of gallons)	22	18.5	19.5	22	24

A gasoline storage tank is filled by a pipeline from a refinery. At the same time, gasoline flows from the tank into trucks that will make deliveries to gasoline stations. The amount of gasoline in the storage tank at time t is given by the twice-differentiable function G , where t is measured in hours and $0 \leq t \leq 10$. Values of $G(t)$, in thousands of gallons, at selected times t are given in the table above. It is known that $G''(t) > 0$ for $5 \leq t < 10$.

- Use the data in the table to estimate the rate of change of the amount of gasoline in the storage tank at time $t = 3$ hours. Show the computations that lead to your answer. Indicate units of measure.
- For $0 < t < 10$, is there a time t at which $G(t)$ is increasing at a rate of 0.2 thousand gallons per hour? Justify your answer.
- It is known that $G'(5) = 0.5$. Use the locally linear approximation for G at time $t = 5$ to approximate the amount of gasoline in the storage tank at time $t = 7$. Is this approximation an overestimate or an underestimate for the actual amount of gasoline in the storage tank at time $t = 7$? Give a reason for your answer.
- The rate at which gasoline flows out of the storage tank into trucks at time t can be modeled by the function R defined by $R(t) = \frac{100t}{t^2+4}$, where t is measured in hours, and $R(t)$ is measured in thousands of gallons. Based on the model, at what time t , for $0 \leq t \leq 10$, is the rate at which gasoline flows out of the storage tank an absolute maximum? Justify your answer.

Part A

Substitution of function values is required. The answer does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct approximation with units.

Solution:

5-5

$G'(3) \approx \frac{G(5)-G(1)}{5-1} = \frac{19.5-18.5}{5-1} = \frac{1}{4} = 0.25$ thousand gallons per hour (which is equivalent to 250 gallons per hour)

Part B

The first point requires identification of the difference quotient. The substitution of function values and conclusion impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{G(10)-G(0)}{10-0}$
- ☐ justification using Mean Value Theorem

Solution:

G is twice differentiable. $\Rightarrow G$ is differentiable. $\Rightarrow G$ is continuous.

$$\frac{G(10)-G(0)}{10-0} = \frac{24-22}{10-0} = \frac{2}{10} = 0.2 \text{ thousand gallons per hour (which is equivalent to 200 gallons per hour)}$$

Therefore, by the Mean Value Theorem, there is a time t , for $0 < t < 10$, at which $G'(t) = 0.2$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation of $G(7)$
- ☐ underestimate with reason

Solution:

$$G(7) \approx G(5) + G'(5)(7-5) = 19.5 + (0.5)(2) = 20.5 \text{ thousand gallons}$$

5-5

$G''(t) > 0$ for $5 \leq t \leq 7$, and so the graph of G is concave up for $5 \leq t \leq 7$. Therefore, the approximation for $G(7)$ using the locally linear approximation for G at time $t = 5$ is an underestimate for the actual amount.

Part D

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $R'(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $R'(t)$
- ☐ sets $R'(t) = 0$
- ☐ identifies $t = 2$ as a candidate
- ☐ answer with justification

Solution:

$$R'(t) = \frac{100(t^2+4) - 100t(2t)}{(t^2+4)^2} = \frac{400 - 100t^2}{(t^2+4)^2} = \frac{100(4-t^2)}{(t^2+4)^2} = 0$$

$$\Rightarrow 4 - t^2 = 0 \Rightarrow t = 2$$

Because $R'(t)$ changes from positive to negative at $t = 2$, R has a relative maximum at $t = 2$. R is increasing on $0 \leq t \leq 2$ and R is decreasing on $2 \leq t \leq 10$. Since $t = 2$ is the only critical point of R on the interval $0 \leq t \leq 10$ that is the location of a relative maximum, it is also the location of the absolute maximum of R on the interval $0 \leq t \leq 10$.

5-5

30. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (years)	0	4	9	12
$R(t)$ (bears per year)	150	156	159	161

The rate at which the number of bears in a region is changing is modeled by the differentiable function R , where $R(t)$ is measured in bears per year and t is measured in years. Selected values of $R(t)$ are shown in the table.

(a) Approximate $R'(7)$ using the average rate of change of R over the interval $4 \leq t \leq 9$. Show the computations that lead to your answer, and indicate units of measure.

(b) Interpret the meaning of $\frac{1}{12} \int_0^{12} R(t) \, dt$ in the context of the problem. Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\frac{1}{12} \int_0^{12} R(t) \, dt$.

(c) The rate at which the number of bears in the region is changing can also be modeled by the function P defined by $P(t) = 150e^{0.02\sqrt{t}}$, where $P(t)$ is measured in bears per year and t is measured in years. There are 300 bears in the region at time $t = 0$. Use this model to find the number of bears in the region at time $t = 12$. Show the setup for your calculations, and give your answer to the nearest whole number of bears.

(d) Using the model defined in part (c), find the time t , for $0 \leq t \leq 12$, at which the number of bears in the region is a maximum. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-5

To earn the first point a response must provide a numerical difference and a quotient. A response of only or earns the first point.

To earn the second point, a response must have “bear per year per year” (or the equivalent) attached to a numerical value.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation with supporting work
- ☐ Units

Solution:

$$R'(7) \approx \frac{R(9) - R(4)}{9 - 4} = \frac{159 - 156}{5} = 0.6$$

$$R'(7) \approx 0.6 \text{ bear per year per year}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must provide “average,” “rate of change,” and “from time $t = 0$ to $t = 12$ ” (or equivalent). Units are not required to earn the first point.

To earn the second point, at least five of the six factors in the Riemann sum must be correct. The factor of $\frac{1}{12}$ is assessed in the third point.

If there is any error in the Riemann sum, the response does not earn the third point.

A response of only $\frac{1}{12}(150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3)$ earns the second and third points.

A response that presents the correct average value with accompanying work that does not explicitly present the six factors and/or the sum process does not earn the second point but will earn the third point. For example, responses of “ $\frac{150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3}{12}$ ” or “ $600, 780, 477 \rightarrow \frac{1857}{12}$ ” would earn the third point but not the second.

A response that presents an answer of $\frac{156 \cdot 4 + 159 \cdot 5 + 161 \cdot 3}{12}$ (which uses a right Riemann sum) earns 1 of the last 2 points. If this response is simplified, the result must be $\frac{1902}{12}$, $\frac{951}{6}$, $\frac{317}{2}$, or 158.5.

An unsupported response of $\frac{1857}{12}$, $\frac{619}{4}$, or 154.75 does not earn the second or third point.

5-5

A response that miscommunicates by equating two unequal quantities, e.g.,

$$\int_0^{12} R(t) \, dt = \frac{150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3}{12}, \text{ does not earn the third point.}$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation
- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$\frac{1}{12} \int_0^{12} R(t) \, dt$ is the average rate of change in the number of bears in the region, in bears per year, from time $t = 0$ to time $t = 12$ years.

$$\begin{aligned} \int_0^{12} R(t) \, dt &\approx R(0) \cdot (4 - 0) + R(4) \cdot (9 - 4) + R(9) \cdot (12 - 9) \\ &= 150 \cdot (4 - 0) + 159 \cdot (9 - 4) + 159 \cdot (12 - 9) \\ &= 150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3 = 1857 \end{aligned}$$

$$\frac{1}{12} \int_0^{12} R(t) \, dt \approx \frac{1857}{12} = \frac{619}{4} = 154.75 \text{ bears per year}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral $\int_a^b P(t) \, dt$ with any limits of integration. If the limits are incorrect, the response does not earn the second point.

The second point cannot be earned without the first point, and is earned only for an answer of 2185 or 2185.339 or 2185.338.

Missing differentials:

- A response of only $\int_0^{12} P(t)$ or $\int_0^{12} P(t) = 1885.3389$ earns the first point but not the second.

5-5

- A response of $300 + \int_0^{12} P(t)$ earns the first point and in the presence of the correct answer earns the second point.
- A response of $\int_0^{12} P(t) + 300$ does not earn the first point but earns the second point for the correct answer.

A response that miscommunicates, e.g., $\int_0^{12} P(t) dt = 1885.3389 + 300$, does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integral
- ☐ Answer

Solution:

$$300 + \int_0^{12} P(t) dt = 300 + 1885.3389 = 2185.3389$$

At $t = 12$ years, there are 2185 bears in the region.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that considers $P = 0$ earns the first point.

A response of “at time $t = 12$ ” with no justification does not earn the second point.

The second point cannot be earned for any answer other than time $t = 12$.

Alternate solution using Candidates Test:

$P(t) = 0$ has no solutions in the interval $0 \leq t \leq 12$, so the maximum number of bears occurs at an endpoint of the interval.

The region has 300 bears at time $t = 0$.

From part (c), the region has 2185 bears at time $t = 12$.

For $0 \leq t \leq 12$, the maximum number of bears is at time $t = 12$ years.

5-5

- The scoring is the same as the model solution.
- A response using this argument may import an incorrect value for the number of bears in the region in year 12 from part (c), provided that value is larger than 300.
- A response using this argument need not report the number of bears at time $t = 12$. Stating that this number is greater than 300 is sufficient.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers sign of P
- ☐ Answer with justification

Solution:

$P(t) > 0$ for all $t \geq 0$, so the bear population is always increasing.

For $0 \leq t \leq 12$, the maximum number of bears is at time $t = 12$ years.

5-5

31. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A particle moves along the x -axis so that its velocity at time t is given by $v(t) = 2t \cdot \cos\left(\frac{t^3}{4}\right)$. The particle is at position $x = 3$ at time $t = 0$.

- (a) For $0 < t < \pi$, find all times t during which the particle is moving to the left. Give a reason for your answer.
- (b) Find the total distance traveled by the particle from time $t = 0$ to time $t = 2$. Show the setup for your calculations.
- (c) Find the acceleration of the particle at time $t = 2$. Show the setup for your calculations. Is the particle speeding up or slowing down at time $t = 2$? Give a reason for your answer.
- (d) Find the rightmost position of the particle for $0 \leq t \leq \pi$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that considers the sign of $v(t)$ earns the first point.

A response that identifies $t = 1.845$ (or $\sqrt[3]{2\pi}$) and/or $t = 2.661$ (or $\sqrt[3]{6\pi}$) and no other values in the interval $0 < t < \pi$ earns the first point.

A response of only $1.845 < t < 2.661$ earns the first point but not the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

5-5

- ☐ Considers $v(t) = 0$
- ☐ Answer with reason

Solution:

The particle is moving to the left when $v(t) < 0$.

$$0 < t < \pi \text{ and } v(t) = 0$$

$$\Rightarrow t = A = 1.845270 \text{ and } t = B = 2.661340$$

The particle is moving left for $1.845 < t < 2.661$ because $v(t) < 0$ in this interval.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for $\int_0^2 |v(t)| dt$ or $\int_0^{1.845} v(t) dt - \int_{1.845}^2 v(t) dt$ or an equivalent expression, with or without the differential.

The second point is earned for an answer of 2.595 regardless of whether the first point was earned.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode,

$$\int_0^2 |v(t)| dt = 3.999391.$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integral
- ☐ Answer

Solution:

$$\int_0^2 |v(t)| dt = 2.595422$$

The total distance traveled by the particle from time $t = 0$ to time $t = 2$ is 2.595.

Part C

5-5

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimum work needed to earn the first point is $v'(2) = -11.744$ (or -11.743).

$a(2) = -11.743863$ is not sufficient to earn the first point. The connection $v'(t) = a(t)$ or $v'(2) = a(2)$ must be explicitly made.

A response that writes an incorrect statement such as $v'(t) = a(2)$ or $dv = a = -11.744$ does not earn the first point but is eligible for the second point. Note that a response that presents $v' = a = -11.744$ at $t = 2$ does earn the first point.

A response does not need to find the value of $v(2)$; the sign is sufficient. However, a response that reports an incorrect sign or value of $v(2)$ does not earn the second point.

If a value for $v(2)$ is presented, that value must be accurate to the number of decimals presented, for zero up to three decimal places.

A response must declare a value for $a(2)$ to be eligible to earn the second point. In order to earn the second point, the presented conclusion must be consistent with a negative velocity at time $t = 2$ and the presented value of $a(2)$.

Degree mode: In degree mode, $v'(2) = 1.991472$ and $v(2) = 3.997563$. A response that declares $v'(2) = 1.991472$ will earn the second point in the presence of $v(2) = 3.997563$ or $v(2) > 0$ and the conclusion that the particle is speeding up.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Acceleration with setup
- ☐ Speeding up with reason

Solution:

$$v'(2) = a(2) = -11.743863$$

The acceleration at time $t = 2$ is -11.744 (or -11.743).

$$v(2) = -1.664587 < 0$$

The particle is speeding up at time $t = 2$ because the particle's velocity and acceleration are both negative at that time.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-5

The first point is earned for an attempt to evaluate $x(A)$.

The second point is earned for considering both 0 and π as candidates, verbally or in a table.

To earn the third point the response must identify the rightmost position at 5.471 and either:

- evaluate x at $t = 0$, $t = A$, $t = B$, and $t = \pi$, or
- evaluate x at $t = A$ and $t = \pi$ and eliminate $t = 0$ and $t = B$ from consideration by arguing that $0 < t < A \Rightarrow v(t) > 0$ and $B < t < \pi \Rightarrow v(t) > 0$.

All evaluations must be correct for the number of digits presented for one up to three decimal places.

Degree mode: The response is only eligible to earn the first two points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x(A)$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

$$0 \leq t \leq \pi \text{ and } v(t) = 0$$

$$\Rightarrow t = 0, t = A = 1.845270, \text{ and } t = B = 2.661340$$

Because $v(t)$ is continuous on $[0, \pi]$, the candidates for the time at which the particle is at its rightmost position are $t = 0$, $t = A$, $t = B$, and $t = \pi$.

$$x(0) = 3$$

$$x(A) = 3 + \int_0^A v(t) dt = 5.471352$$

$$x(B) = 3 + \int_0^B v(t) dt = 3.151377$$

$$x(\pi) = 3 + \int_0^\pi v(t) dt = 4.972567$$

The rightmost position of the particle for $0 \leq t \leq \pi$ is $x = 5.471$.

5-5

32. Particle X moves along the positive x -axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$.
- (a) Is particle X moving toward the left or toward the right at time $t = 1$? Give a reason for your answer.
- (b) At what time $t \geq 0$ is particle X farthest to the left? Justify your answer.
- (c) A second particle, Y , moves along the positive y -axis so that its position at time t is given by $y(t) = 7t + 3$. At any time t , $t \geq 0$, the origin and the positions of the particles X and Y are the vertices of a triangle in the first quadrant. Find the rate of change of the area of the triangle at time $t = 1$. Show the work that leads to your answer.

Part A

The response can earn up to 2 points:

1 point: For consideration of $x'(1)$

1 point: For correct answer with reason

$$x'(t) = 15t^2 - 18t$$

$$x'(1) = 15 - 18 = -3$$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$.



0	1	2
---	---	---

The response earns all of the following points:

1 point: For consideration of $x'(1)$

1 point: For correct answer with reason

$$x'(t) = 15t^2 - 18t$$

$$x'(1) = 15 - 18 = -3$$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$.

Part B

The response can earn up to 3 points:

1 point: For consideration of $x'(t) = 0$

1 point: For identification of $t = \frac{6}{5}$

5-5

1 point: For correct answer with reason

$$x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0, t = \frac{6}{5}$$

Since $x'(t) < 0$, for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$, the particle is farthest to the left at time $t = \frac{6}{5}$.



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: For consideration of $x'(t) = 0$

1 point: For identification of $t = \frac{6}{5}$

1 point: For correct answer with reason

$$x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0, t = \frac{6}{5}$$

Since $x'(t) < 0$, for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$, the particle is farthest to the left at time $t = \frac{6}{5}$.

Part C

The response can earn up to 4 points:

1 point: For the area function

$$\begin{aligned} \text{Area} = A(t) &= \frac{1}{2}x(t)y(t) \\ &= \frac{1}{2}(5t^3 - 9t^2 + 7)(7t + 3) \end{aligned}$$

2 points: For the derivative

$$A'(t) = \frac{1}{2}[(15t^2 - 18t)(7t + 3) + (5t^3 - 9t^2 + 6)(7)]$$

1 point: For the correct answer

$$A'(1) = \frac{1}{2}[(-3)(10) + (3)(7)] = \frac{1}{2}[-30 + 21] = -\frac{9}{2}$$



0	1	2	3	4
---	---	---	---	---

The response earns all of the following points:

5-5

1 point: For the area function

$$\begin{aligned}\text{Area} &= A(t) = \frac{1}{2}x(t)y(t) \\ &= \frac{1}{2}(5t^3 - 9t^2 + 7)(7t + 3)\end{aligned}$$

2 points: For the derivative

$$A'(t) = \frac{1}{2}[(15t^2 - 18t)(7t + 3) + (5t^3 - 9t^2 + 6)(7)]$$

1 point: For the correct answer

$$A'(1) = \frac{1}{2}[(-3)(10) + (3)(7)] = \frac{1}{2}[-30 + 21] = -\frac{9}{2}$$

5-5

33.  The penguin population on an island is modeled by a differentiable function P of time t , where $P(t)$ is the number of penguins and t is measured in years, for $0 \leq t \leq 40$. There are 100,000 penguins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$$B(t) = 1000e^{0.06t} \text{ penguins per year}$$

and the death rate for the penguins on the island is modeled by

$$D(t) = 250e^{0.1t} \text{ penguins per year.}$$

- (a) What is the rate of change of the penguin population on the island at time $t = 0$?
- (b) To the nearest whole number, what is the penguin population on the island at time $t = 40$?
- (c) To the nearest whole number, what is the average rate of change of the penguin population on the island for $0 \leq t \leq 40$?
- (d) To the nearest whole number, find the absolute minimum penguin population and the absolute maximum penguin population on the island for $0 \leq t \leq 40$. Show the analysis that leads to your answer.

Part A

The response can earn up to 1 point:

1 point is earned for the answer

$$P'(0) = B(0) - D(0) = 1000 - 250 = 750 \text{ penguins per year}$$



0	1
---	---

The response earns all of the following point:

1 point is earned for the answer

$$P'(0) = B(0) - D(0) = 1000 - 250 = 750 \text{ penguins per year}$$

Part B

The response can earn up to 2 points:

1 point is earned for correct limits

1 point is earned for the integrand

1 point is earned for answer

$$P(40) = 100000 + \int_0^{40} (B(t) - D(t)) dt$$

$$= 100000 + 33057.56459$$

5-5

There are 133,058 penguins on the island.



0	1	2	3
---	---	---	---

The student earns all of the following point:

1 point is earned for correct limits

1 point is earned for the integrand

1 point is earned for answer

$$P(40) = 100000 + \int_0^{40} (B(t) - D(t)) dt$$

$$= 100000 + 33057.56459$$

There are 133,058 penguins on the island.

Part C

The response can earn up to 1 point:

1 point is earned for the answer

$$\frac{1}{40} \int_0^{40} (B(t) - D(t)) dt = 826.439$$

OR

$$\frac{P(40) - P(0)}{40 - 0} = \frac{133058 - 100000}{40} = 826.45$$

The average rate of change is 826 penguins per year.



0	1
---	---

The student earns all of the following point:

1 point is earned for the answer

$$\frac{1}{40} \int_0^{40} (B(t) - D(t)) dt = 826.439$$

OR

5-5

$$\frac{P(40) - P(0)}{40 - 0} = \frac{133058 - 100000}{40} = 826.45$$

The average rate of change is 826 penguins per year.

Part D

The response can earn up to 4 points:

1 point is earned for the expression $B(t) - D(t) = 0$

$$B(t) - D(t) = 0$$

1 point is earned for the solution for t

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point is earned for the minimum value

1 point is earned for the maximum value

$$P(0) = 100000$$

$$P(A) = 100000 + \int_0^A (B(t) - D(t)) dt = 139166.667$$

$$P(40) = 133058$$

The minimum population is 100,000 and the maximum population is 139,167 penguins.



0	1	2	3	4
---	---	---	---	---

The student earns all of the following point:

1 point is earned for the expression $B(t) - D(t) = 0$

$$B(t) - D(t) = 0$$

1 point is earned for the solution for t

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point is earned for the minimum value

5-5

1 point is earned for the maximum value

$$P(0) = 100000$$

$$P(A) = 100000 + \int_0^A (B(t) - D(t)) dt = 139166.667$$

$$P(40) = 133058$$

The minimum population is 100,000 and the maximum population is 139,167 penguins.

34. Let f be the function given by $f(x) = -x^3 + 3x^2 + 24x$. What is the absolute maximum value of f on the closed interval $[-6, 6]$?

(A) -6

(B) 36

(C) 80

(D) 180



Answer D

Correct. The absolute maximum on a closed interval will occur at a critical point or an endpoint. Since

$f'(x) = -3x^2 + 6x + 24 = -3(x^2 - 2x - 8) = -3(x + 2)(x - 4)$, the critical points are at $x = -2$ and $x = 4$.

$$f(-6) = 180$$

$$f(-2) = -28$$

$$f(4) = 80$$

$$f(6) = 36$$

Therefore, the absolute maximum value is 180.

5-5

35. Let g be the function given by $g(x) = x^4 - \frac{4}{3}x^3$. At what value of x on the closed interval $[-1, 1]$ does g have an absolute maximum?

(A) -1 (B) 0 (C) 1 (D) $\frac{4}{3}$ **Answer A**

Correct. The absolute maximum on a closed interval will occur at a critical point or an endpoint. Since $g'(x) = 4x^3 - 4x^2 = 4x^2(x - 1)$, the critical points are at $x = 0$ and $x = 1$. The endpoints are at $x = -1$ and $x = 1$.

$$g(-1) = \frac{7}{3}$$

$$g(0) = 0$$

$$g(1) = -\frac{1}{3}$$

The absolute maximum occurs at $x = -1$ where $g(-1) = \frac{7}{3}$.

36. Let f be the function defined by $f(x) = 12x - x^3$. What is the absolute minimum value of f on the closed interval $[0, 3]$?

(A) -16 (B) 0 (C) 9 (D) 16 **Answer B**

Correct. Because f is continuous on the interval $[0, 3]$, it has an absolute minimum value on $[0, 3]$ by the Extreme Value Theorem. The derivative of f is given by $f'(x) = 12 - 3x^2 = 3(2 - x)(2 + x)$. Therefore, f has critical points at $x = -2$ and $x = 2$, the two values of x at which $f'(x) = 0$. Because absolute extrema of a function on a closed interval can only occur at critical points or at endpoints, the candidates for the location of the absolute minimum are $x = 0$, $x = 2$, and $x = 3$. Note that $x = -2$ is not in the interval $[0, 3]$. Because $f(0) = 0$, $f(2) = 16$, and $f(3) = 9$, the absolute minimum value of f on the interval $[0, 3]$ is 0.

5-5

37. Let g be the function defined by $g(x) = |x| + 2|1 - x|$. What is the absolute minimum value of g on the closed interval $[-2, 2]$?

- (A) 1
(B) 2
(C) 4
(D) 8

**Answer A**

Correct. Because g is continuous on the interval $[-2, 2]$, it has an absolute minimum value on $[-2, 2]$ by the Extreme Value Theorem. g has critical points at $x = 0$ and $x = 1$, the two values of x at which the derivative g' does not exist. Because absolute extrema of a function on a closed interval can only occur at critical points or at endpoints, the candidates for the location of the absolute minimum are $x = -2$, $x = 0$, $x = 1$, and $x = 2$. Because $g(-2) = 8$, $g(0) = 2$, $g(1) = 1$, and $g(2) = 4$, the absolute minimum value of g on the interval $[-2, 2]$ is 1.

38. Let g be the function defined by $g(x) = (x^2 - x + 1)e^x$. What is the absolute maximum value of g on the interval $[-4, 1]$?

- (A) 1
(B) e
(C) $\frac{3}{e}$
(D) $\frac{21}{e^4}$

**Answer B**

Correct. Because g is continuous on the interval $[-4, 1]$, it has an absolute maximum value on $[-4, 1]$ by the Extreme Value Theorem. The derivative of g is given by $g'(x) = (2x - 1)e^x + (x^2 - x + 1)e^x = x(x + 1)e^x$. Therefore, g has critical points at $x = -1$ and $x = 0$, the two values of x at which $g'(x) = 0$. Because absolute extrema of a function on a closed interval can occur only at critical points or at endpoints, the candidates for the location of the absolute maximum are $x = -4$, $x = -1$, $x = 0$, and $x = 1$. Because $g(-4) = 21e^{-4}$, $g(-1) = 3e^{-1}$, $g(0) = 1$, and $g(1) = e$, the absolute maximum value of g on the interval $[-4, 1]$ is e .

5-5

39. Let f be the function defined by $f(x) = x^3 - \frac{3}{2}x^2 - 6x$. What is the absolute maximum value of f on the interval $[-2, 3]$?
- (A) -10
- (B) $-\frac{9}{2}$
- (C) $\frac{7}{2}$ ✓
- (D) 3

Answer C

Correct. Because f is continuous on the interval $[-2, 3]$, it has an absolute maximum value on $[-2, 3]$ by the Extreme Value Theorem. The derivative of f is given by $f'(x) = 3x^2 - 3x - 6$. Therefore, f has critical points at $x = -1$ and $x = 2$, the two values of x in the interval $[-2, 3]$ at which $f'(x) = 0$. Because absolute extrema of a function on a closed interval can only occur at critical points or at endpoints, the candidates for the location of the absolute maximum are $x = -2$, $x = -1$, $x = 2$, and $x = 3$.

$$f(-2) = -2$$

$$f(-1) = \frac{7}{2}$$

$$f(2) = -10$$

$$f(3) = -\frac{9}{2}$$

Therefore, the absolute maximum value of f on the interval $[-2, 3]$ is $\frac{7}{2}$.

5-5

40. Let f be the function defined by $f(x) = x \cos x - \sin x$. What is the absolute maximum value of f on the interval $[-\frac{\pi}{2}, 2\pi]$?
- (A) $-\pi$
- (B) 2π ✓
- (C) 0
- (D) 1

Answer B

Correct. Because f is continuous on the interval $[-\frac{\pi}{2}, 2\pi]$, it has an absolute maximum value on $[-\frac{\pi}{2}, 2\pi]$ by the Extreme Value Theorem. The derivative of f is given by $f'(x) = -x \sin x$. Therefore, f has critical points at $x = 0$, $x = \pi$, and $x = 2\pi$ in the interval, the three values of x at which $f'(x) = 0$. Because absolute extrema of a continuous function on a closed interval can occur only at critical points or at endpoints, the candidates for the location of the absolute maximum are $x = -\frac{\pi}{2}$, $x = 0$, $x = \pi$, and $x = 2\pi$.

$$f\left(-\frac{\pi}{2}\right) = 1$$

$$f(0) = 0$$

$$f(\pi) = -\pi$$

$$f(2\pi) = 2\pi$$

Therefore, the absolute maximum value of f on the interval $[-\frac{\pi}{2}, 2\pi]$ is 2π .

41. Let g be the function given by $g(x) = 3x^4 - 8x^3$. At what value of x on the closed interval $[-2, 2]$ does g have an absolute maximum?
- (A) -2 ✓
- (B) 0
- (C) 2
- (D) $\frac{8}{3}$

Answer A

Correct. The absolute maximum on a closed interval will occur at a critical point or an endpoint. Since $g'(x) = 12x^3 - 24x^2$, the critical points are at $x = 0$ and $x = 2$. The endpoints are at $x = -2$ and $x = 2$.

$$g(-2) = 112$$

$$g(0) = 0$$

$$g(2) = -16$$

The absolute maximum occurs at $x = -2$ where $g(-2) = 112$.

5-5

42. Let f be the function defined by $f(x) = 3x^2 - x^3$. What is the absolute minimum value of f on the closed interval $[1, \frac{5}{2}]$?
- (A) 0
- (B) 2
- (C) $\frac{25}{8}$
- (D) 4

Answer B

Correct. Because f is continuous on the interval $[1, \frac{5}{2}]$, it has an absolute minimum value on $[1, \frac{5}{2}]$ by the Extreme Value Theorem. The derivative of f is given by $f'(x) = 6x - 3x^2 = 3x(2 - x)$. Therefore, f has critical points at $x = 0$ and $x = 2$, the two values of x at which $f'(x) = 0$. Because absolute extrema of a function on a closed interval can only occur at critical points or at endpoints, the candidates for the location of the absolute minimum are $x = 1$, $x = 2$, and $x = \frac{5}{2}$. Note that $x = 0$ is not in the interval $[1, \frac{5}{2}]$. Because $f(1) = 2$, $f(2) = 4$, and $f(\frac{5}{2}) = \frac{25}{8}$, the absolute minimum value of f on the interval $[1, \frac{5}{2}]$ is 2.

43. Let g be the function defined by $g(x) = |x| - 3|x + 1|$. What is the absolute maximum value of g on the closed interval $[-2, 2]$?
- (A) 1
- (B) -1
- (C) -3
- (D) -7

Answer A

Correct. The derivative of g is given by $g'(x) = \frac{x}{|x|} - \frac{3(x+1)}{|x+1|}$. The critical points of g are $x = 0$ and $x = -1$ because g' does not exist at those two values of x . Because g is continuous on the interval $[-2, 2]$, it has an absolute maximum value on $[-2, 2]$ by the Extreme Value Theorem. Because absolute extrema of a function on a closed interval can occur only at critical points or at endpoints, the candidates for the location of the absolute maximum are $x = -2$, $x = -1$, $x = 0$, and $x = 2$.

$$g(-2) = -1$$

$$g(-1) = 1$$

$$g(0) = -3$$

$$g(2) = -7$$

Therefore, the absolute maximum value of g on the interval $[-2, 2]$ is 1.

5-5

44. Let f be the function defined by $f(x) = \sin x + \cos x$. What is the absolute minimum value of f on the interval $[0, 2\pi]$?
- (A) -2
- (B) $-\sqrt{2}$ ✓
- (C) -1
- (D) 0

Answer B

Correct. Because f is continuous on the interval $[0, 2\pi]$, it has an absolute minimum value on $[0, 2\pi]$ by the Extreme Value Theorem. The derivative of f is given by $f'(x) = \cos x - \sin x$. Therefore, f has critical points at $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$, the two values of x in the interval $[0, 2\pi]$ at which $f'(x) = 0$. Because absolute extrema of a function on a closed interval can only occur at critical points or at endpoints, the candidates for the location of the absolute minimum are $x = 0$, $x = \frac{\pi}{4}$, $x = \frac{5\pi}{4}$, and $x = 2\pi$. Because $f(0) = 1$, $f(\frac{\pi}{4}) = \sqrt{2}$, $f(\frac{5\pi}{4}) = -\sqrt{2}$, and $f(2\pi) = 1$, the absolute minimum value of f on the interval $[0, 2\pi]$ is $-\sqrt{2}$.

45. Let f be the function given by $f(x) = 2x^3 + 3x^2 + 1$. What is the absolute maximum value of f on the closed interval $[-3, 1]$?
- (A) 1
- (B) 2
- (C) 6 ✓
- (D) 26

Answer C

Correct. The absolute maximum on a closed interval will occur at a critical point or an endpoint. Since $f'(x) = 6x^2 + 6x = 6x(x + 1)$, the critical points are at $x = -1$ and $x = 0$.

$f(-3) = -26$

$f(-1) = 2$

$f(0) = 1$

$f(1) = 6$

Therefore, the absolute maximum value is 6.

5-5

46. The function g is differentiable and satisfies $g(-1) = 4$ and $g'(-1) = 2$. What is the approximation of $g(-1.2)$ using the line tangent to the graph of g at $x = -1$?

(A) 3.6



(B) 3.8

(C) 4.2

(D) 4.4

Answer A

Correct. An equation of the line tangent to the graph of g at $x = a$ is $y = g(a) + g'(a)(x - a)$. In this question, $a = -1$. The value of y when $x = -1.2$ would be an approximation to $g(-1.2)$.

$$g(-1.2) \approx g(-1) + g'(-1)(-1.2 - (-1)) = 4 + 2(-0.2) = 3.6$$